Estimating Vegetation Change from Fossil Pollen to Planetary Boundaries

A Global, Long-Term Synthesis Using Big Data

Ondřej Mottl

TIBS 2026

Scan the QR code to follow along!

6/9 global Earth Planetary Boundaries are already exceeded

(Richardson et al. 2023) *Science Advances*

Image generated by Bing

~20%

*Species to be exposed to **dangerous temperature** condition (with global increase of 2.0°C)*

IPCC AR6 Synthesis Report

Image by Justus Menke from Pexels

What if …

… we inform the conservation policies by incorporating a global, long-term synthesis on ecosystem dynamics using paleoecological Big Data?

🙋 Hi! I am Ondřej Mottl

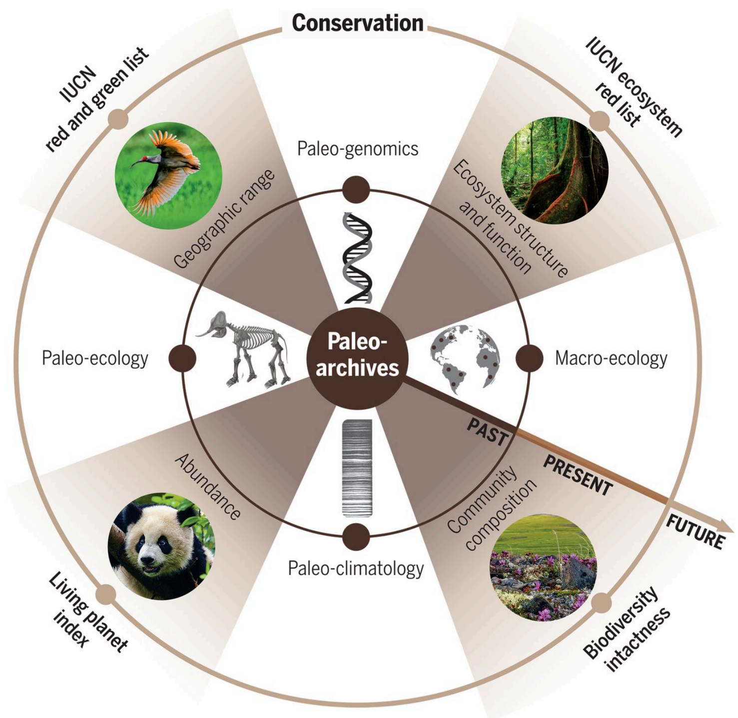
Assistant Professor at 🏛️ Charles University, Prague, CZ

Interested in macroecology, palaeoecology, biodiversity, and data science

Head of the 🧑‍💻 Laboratory of Quantitative Ecology

- 📧 ondrej.mottl(at)natur.cuni.cz
- 🦋 ondrejmottl.bsky.social
- 🐱 Github
- ORCID
- 🌐 Personal webpage

Palaeoecology



Science

HOME > SCIENCE > VOL. 369, NO. 6507 > USING PALEO-ARCHIVES TO SAFEGUARD BIODIVERSITY UNDER CLIMATE CHANGE

Using paleo-archives to safeguard biodiversity under climate change

DAMIEN A. FORDHAM, STEPHEN T. JACKSON, STUART C. BROWN, BRIAN HUNTLEY, BARRY W. BROOK, DORTHE DAHL-JENSEN, M. THOMAS P. GILBERT, BETTE L. OTTO-BLIESNER, ANDERS SVENSSON, [...], AND DAVID NOGUES-BRAVO +8 authors Authors Info & Affiliations

SCIENCE • 28 Aug 2020 • Vol 369, Issue 6507 • DOI: 10.1126/science.abc5654

Because warning signals … are commonly identified by using time-series … data, paleo-archives provide opportunities to test conservation criteria …

Fordham et al. (2020) *Science*

Fossil pollen as a window to the past vegetation patterns

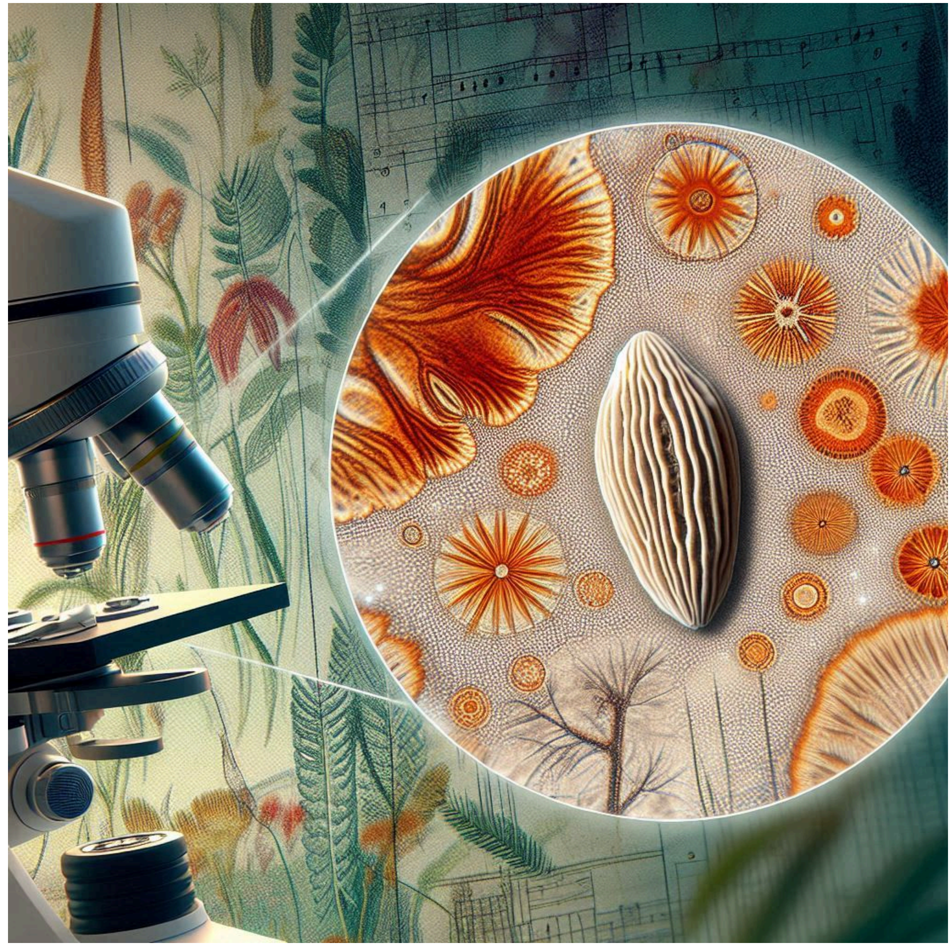


Image created by Bing

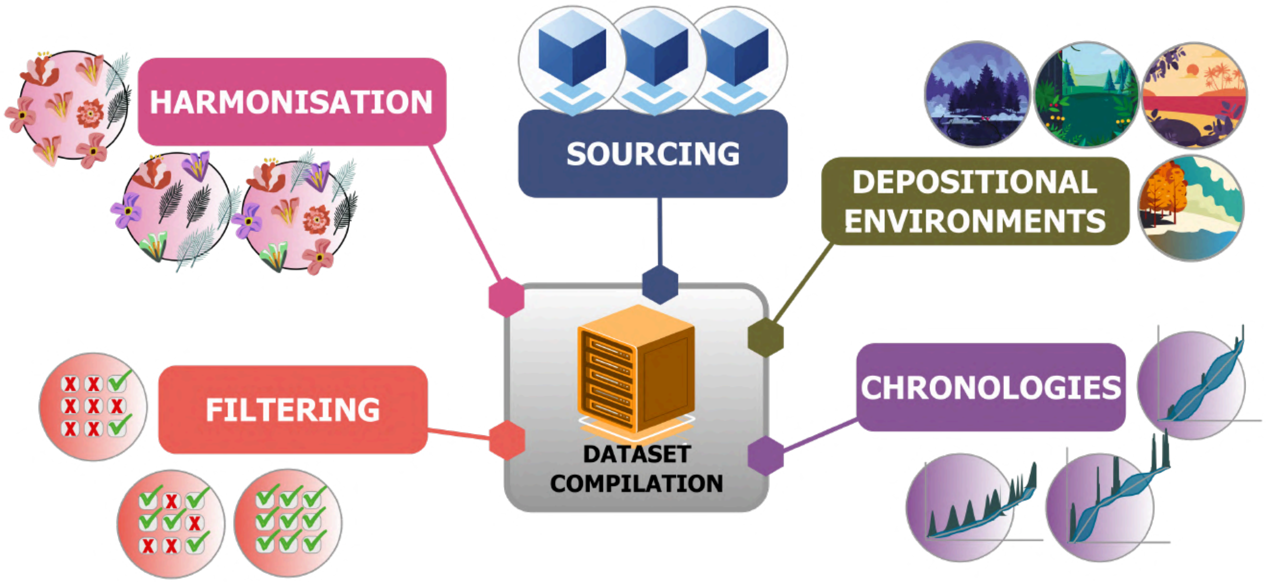
Neotoma Paleoecology Database

>6650

fossil pollen timeseries (records)

Fossil pollen datasets accessed from Neotoma on 2025-12-09

Palaeoecological dataset compilation



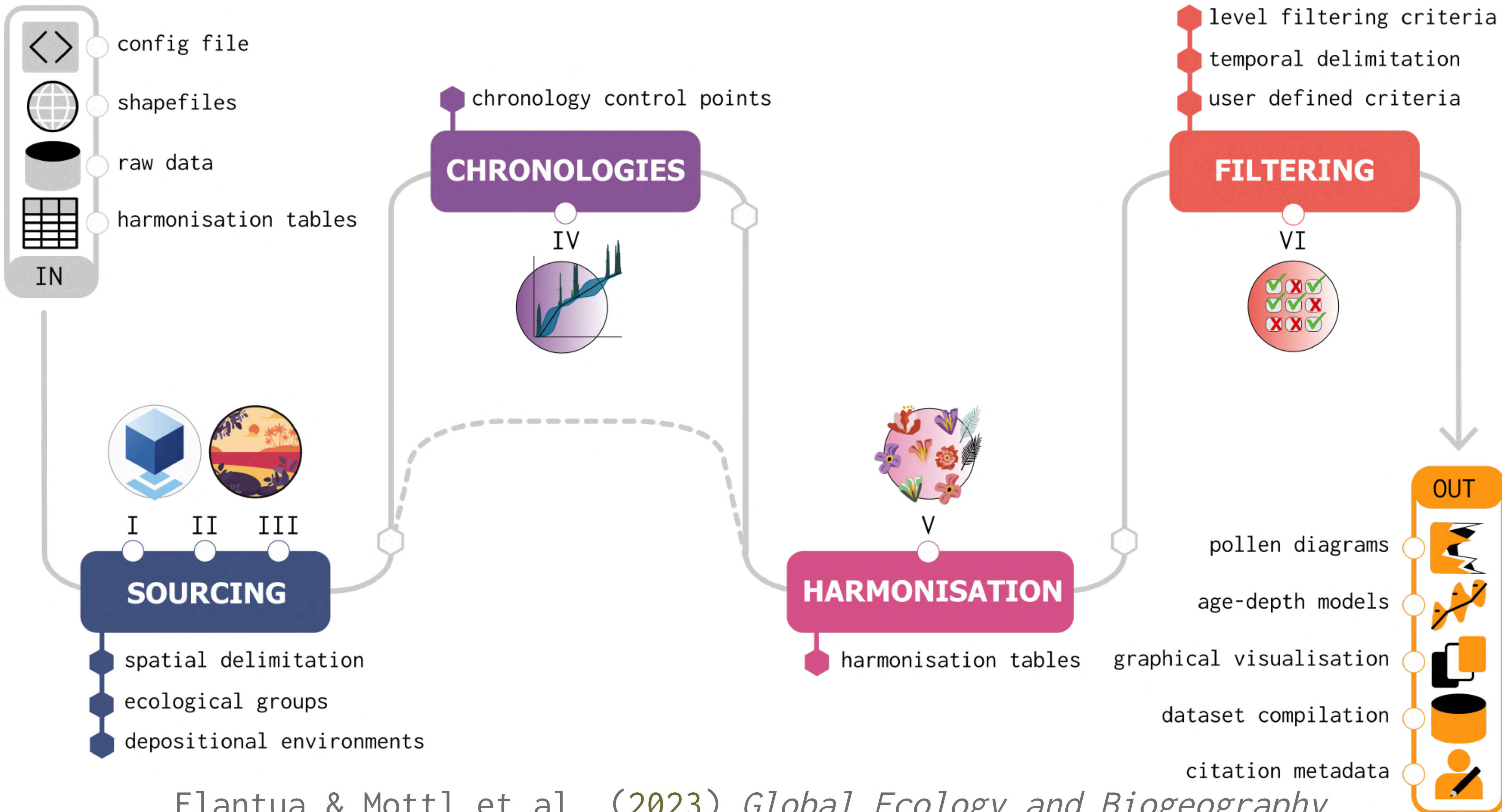
- Macro-scale data analysis
- Project specific
- Data quality control and reproducibility

Adjusted from Flantua & Mottl et al. (2023) *Global Ecology and Biogeography*

The workflow to process **global** palaeoecological data of **fossil pollen** for **vegetation**-based **macroecological** **synthesis**

Flantua & Mottl et al. (2023) *Global Ecology and Biogeography*

FOSSILPOL workflow



Flantua & Mottl et al. (2023) *Global Ecology and Biogeography*

Documentation (website)

PUBLISHED
February 22, 2022

MODIFIED
March 12, 2025

FOSSILPOL project

Summary

This website presents all information needed about the FOSSILPOL workflow that aims to process and standardise global palaeoecological pollen data.

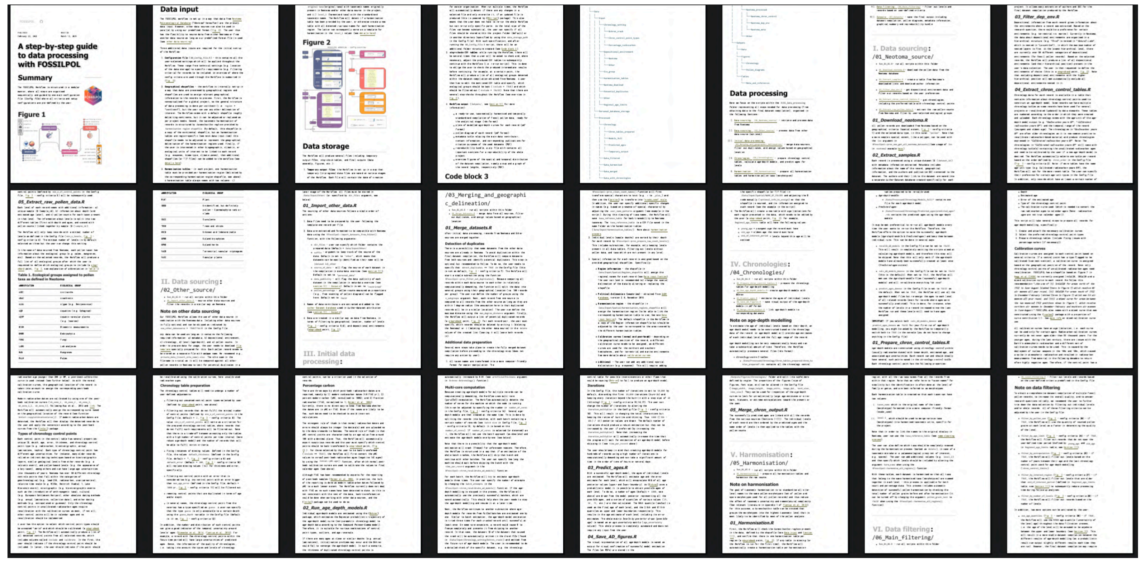
Authors

The FOSSILPOL workflow has been developed during the ERC project called the *Humans on Planet Earth (HOPE)* team at the University of Bergen:

Suzette G.A. Flantua*, Ondrej Mottl*, Vivian Felde*, Kuber P. Bhatta*, Hilary Birks, John-Arvid Grytnes, Alistair Seddon, H. John B. Birks

* shared first authors

Ondřej Mottl is the main maintainer of the FOSSILPOL project



Now bilingual! (English/Spanish)

Check it out! - bit.ly/FOSSILPOL

Flantua & Mottl et al. (2023) *Global Ecology and Biogeography*

FOSSILPOL

- Open-source and reproducible workflow written in R
- Handles all the steps and flags places, where the user needs to specify their choices
- Modular structure to allow flexibility and customisation for different projects

Global Ecology and Biogeography
A Journal of Macroecology

METHOD | Open Access

A guide to the processing and standardization of global palaeoecological data for large-scale syntheses using fossil pollen

Suzette G. A. Flantua, Ondrej Mottl, Vivian A. Felde, Kuber P. Bhatta, Hilary H. Birks, John-Arvid Grytnes, Alistair W. R. Seddon, H. John B. Birks

First published: 02 May 2023 | https://doi.org/10.1111/geb.13693

FOSSILPOL

- Open-source and reproducible workflow written in R
- Handles all the steps and flags places, where the user needs to specify their choices
- Modular structure to allow flexibility and customisation for different projects

Global Ecology and Biogeography
A Journal of Macroecology

METHOD | Open Access

A guide to the processing and standardization of global palaeoecological data for large-scale syntheses using fossil pollen

Suzette G. A. Flantua, Ondrej Mottl, Vivian A. Felde, Kuber P. Bhatta, Hilary H. Birks, John-Arvid Grytnes, Alistair W. R. Seddon, H. John B. Birks

First published: 02 May 2023 | https://doi.org/10.1111/geb.13693

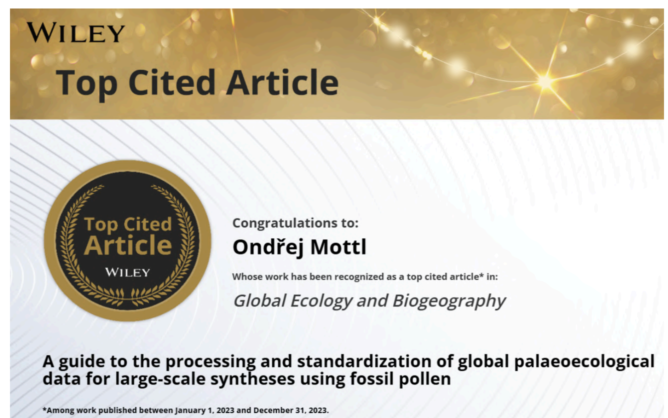
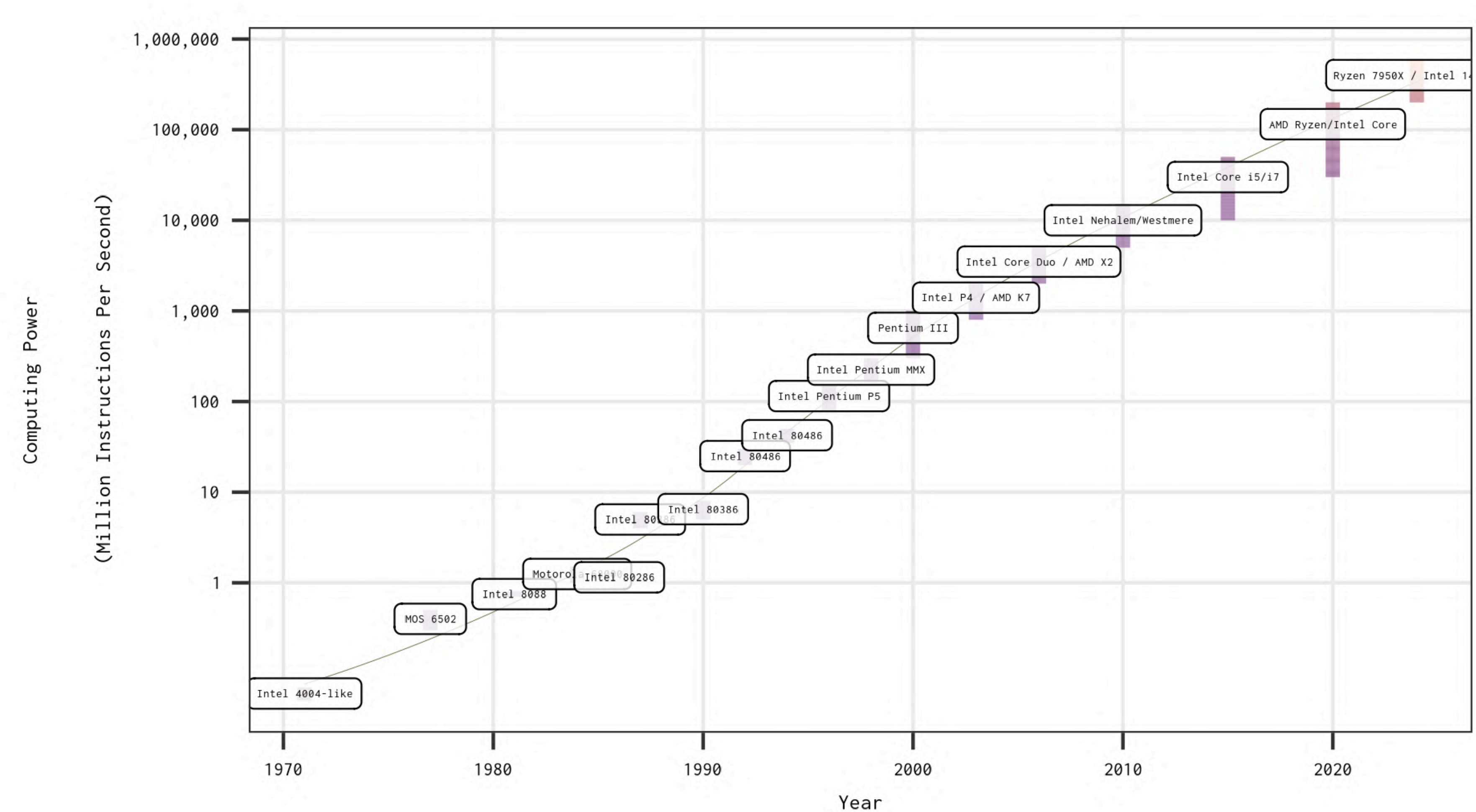


Image by Johannes Plenio from Pexels

The power of a personal computer



The data obtained from [wikipedia/Instructions_per_second]()

Analytical palaeoecology

Challenges and directions in analytical paleobiology

Published online by Cambridge University Press: **27 February 2023**

Erin M. Dillon, Emma M. Dunne, Tom M. Womack, Miranta Kouvari, Ekaterina Larina, Jordan Ray Claytor, Angelina Ivkić, Mark Juhn, Pablo S. Milla Carmona and Selina Viktor Robson ...Show all authors

Advancements

Best practices and reproducible workflows for data standardization, Analyses that can accomodate heterogeneous datasets, Incentive structures that reward software development, Awareness of sampling context and biases, Best practices and conceptual frameworks to guide data integration, Development of taxon-free metrics, Training in quantitative methods and Data Science, Collaboration with data scientists, Interdisciplinary communication, collaboration, and funding

Dillon et al. (2023) *Paleobiology*

Analytical palaeoecology

Challenges and directions in analytical paleobiology

Published online by Cambridge University Press: **27 February 2023**

Erin M. Dillon, Emma M. Dunne, Tom M. Womack, Miranta Kouvari, Ekaterina Larina, Jordan Ray Claytor, Angelina Ivkić, Mark Juhn, Pablo S. Milla Carmona and Selina Viktor Robson ...Show all authors

Advancements

Best practices and reproducible workflows for data standardization, Analyses that can accomodate heterogeneous datasets, Incentive structures that reward software development, Awareness of sampling context and biases, Best practices and conceptual frameworks to guide data integration, Development of taxon-free metrics, Training in quantitative methods and Data Science, Collaboration with data scientists, Interdisciplinary communication, collaboration, and funding

Erin Dillon has a talk directly after me!

Dillon et al. (2023) *Paleobiology*

Statistical advancements

New methods accounting for effort, bias, and uncertainty

Photo by Andrea Piacquadio from Pexels

Statistical advancements

New methods accounting for effort, bias, and uncertainty

- Hierarchical regression models
- Joined Species Distribution Models (jSDM)
- Model-based ordinations
- Copula methods
- …

Photo by Andrea Piacquadio from Pexels

Statistical advancements

New methods accounting for effort, bias, and uncertainty

- Hierarchical regression models
- Joined Species Distribution Models (jSDM)
- Model-based ordinations
- Copula methods
- …

More details in a moment in talk by Gavin Simpson in this Symposium!

Photo by Andrea Piacquadio from Pexels

Estimating Rate of Change (RoC)

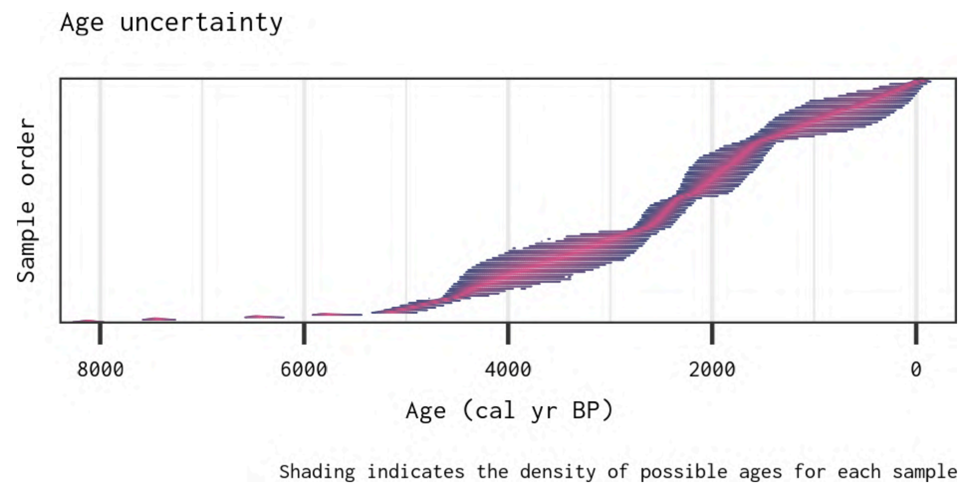
Issues with palaeoecological RoC estimation

Estimating Rate of Change (RoC)

Issues with palaeoecological RoC estimation

- Age uncertainties

example data



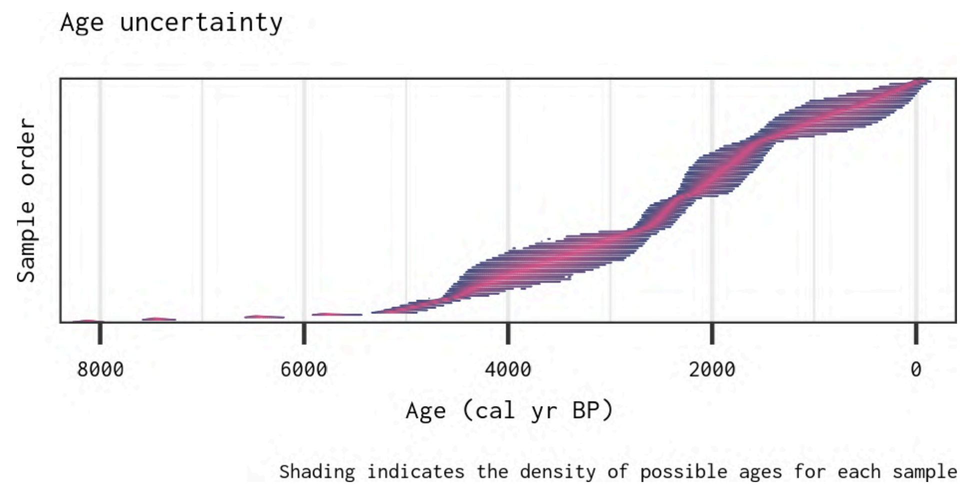
Estimating Rate of Change (RoC)

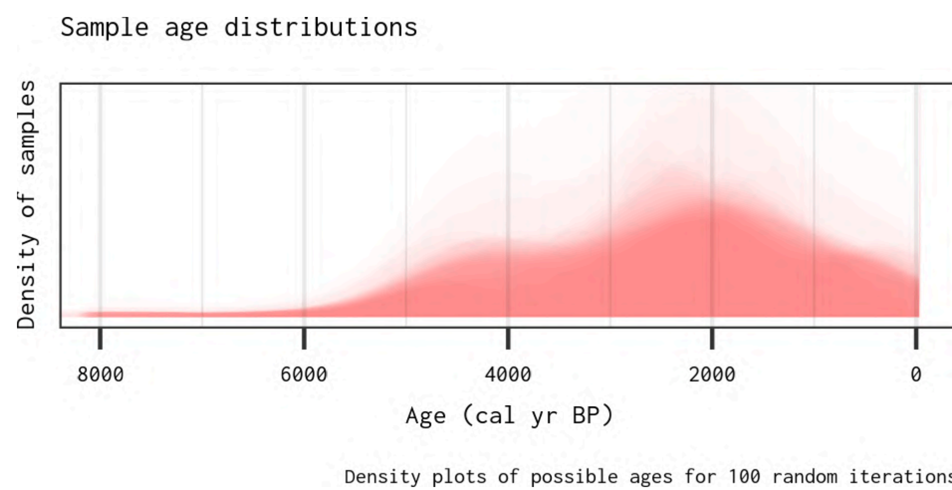
Issues with palaeoecological RoC estimation

- Age uncertainties

- Uneven temporal distribution of samples

example data





Estimating Rate of Change (RoC)

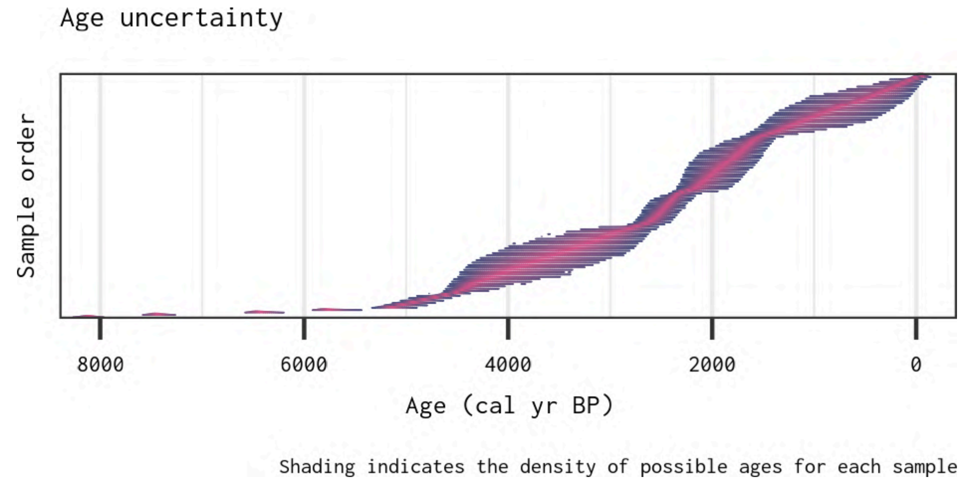
Issues with palaeoecological RoC estimation

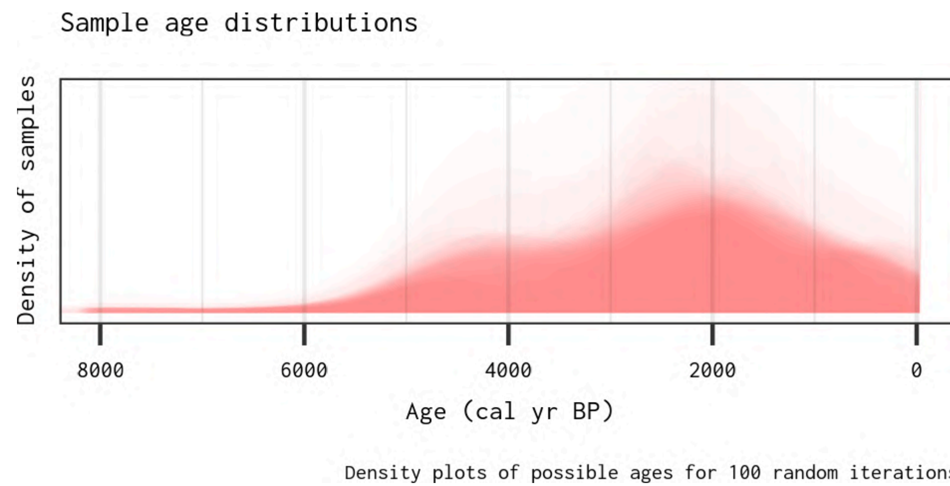
- Age uncertainties

- Uneven temporal distribution of samples

- Taxonomic uncertainties

example data





{RRatepol}

An R package for estimating rate of change (RoC) from multivariate (community) data in time series

Mottl et al. (2021) *Review of Palaeobotany and Palynology*

{RRatepol}

An R package for estimating rate of change (RoC) from multivariate (community) data in time series

- Handles uneven time series

Mottl et al. (2021) *Review of Palaeobotany and Palynology*

{RRatepol}

An R package for estimating rate of change (RoC) from multivariate (community) data in time series

- Handles uneven time series
- Propagates uncertainty (age, taxonomic)

Mottl et al. (2021) *Review of Palaeobotany and Palynology*

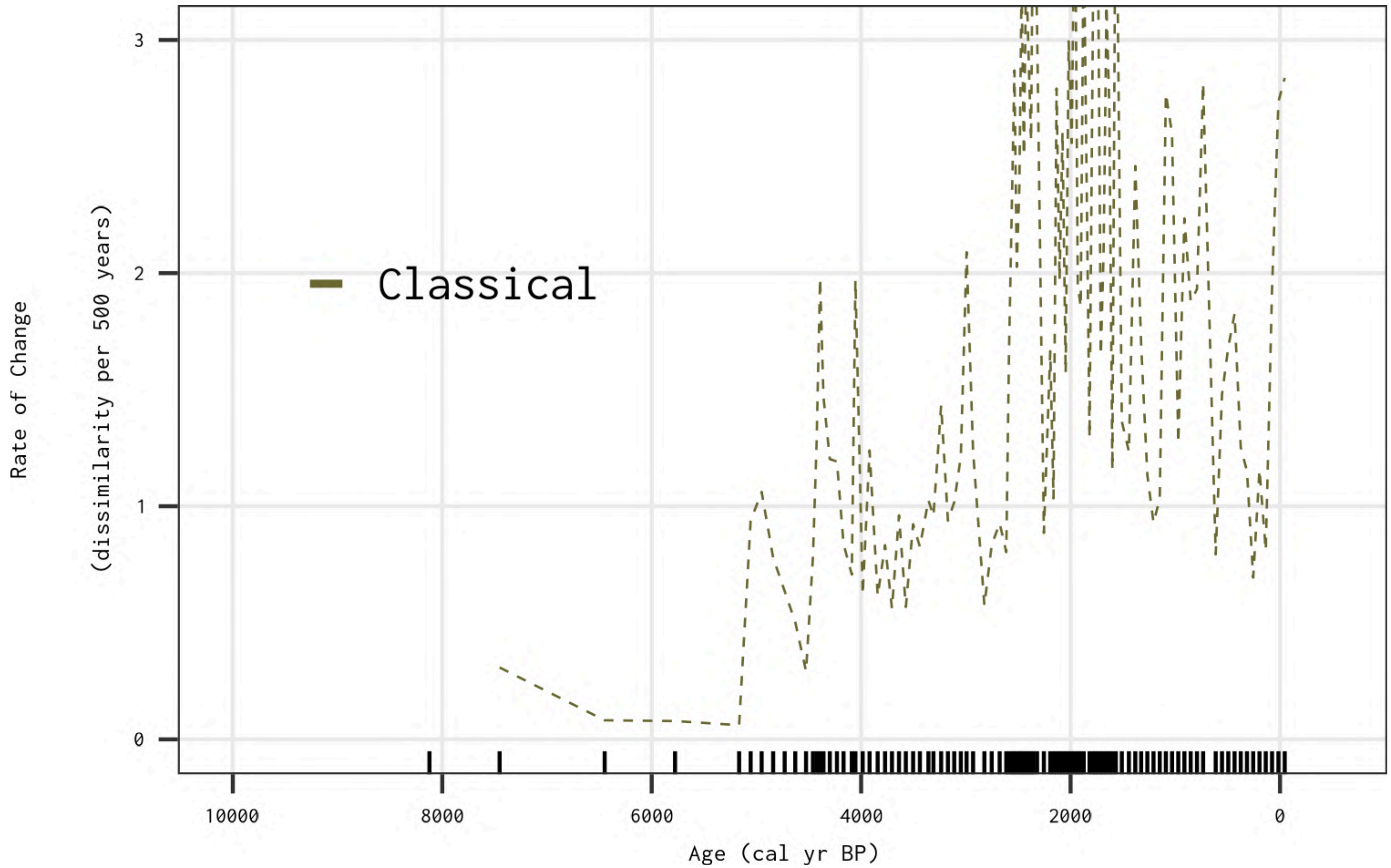
{RRatepol}

An R package for estimating rate of change (RoC) from multivariate (community) data in time series

- Handles uneven time series
- Propagates uncertainty (age, taxonomic)
- Can work with various data types (palaeoecological proxies)

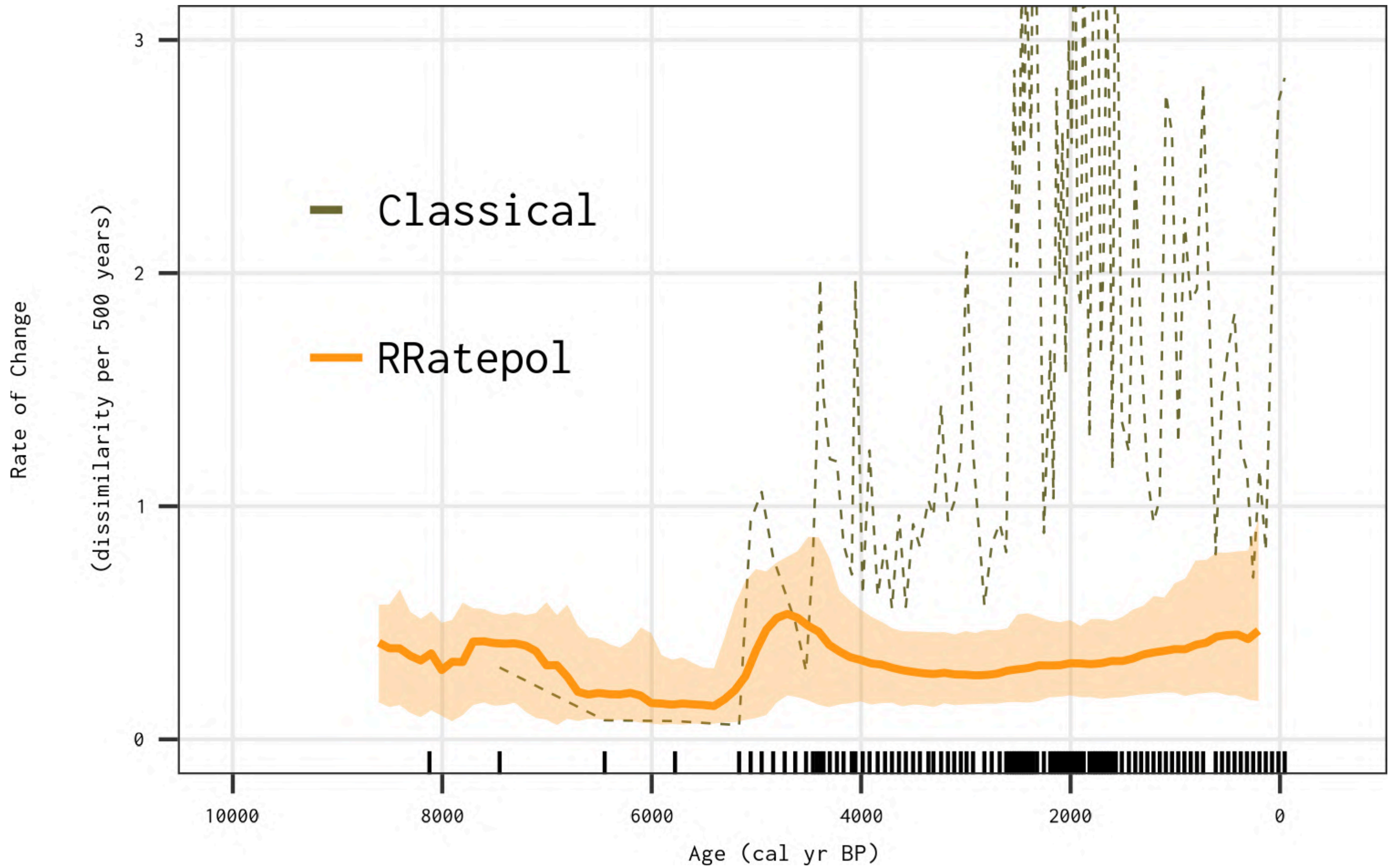
Mottl et al. (2021) *Review of Palaeobotany and Palynology*

{RRatepol}



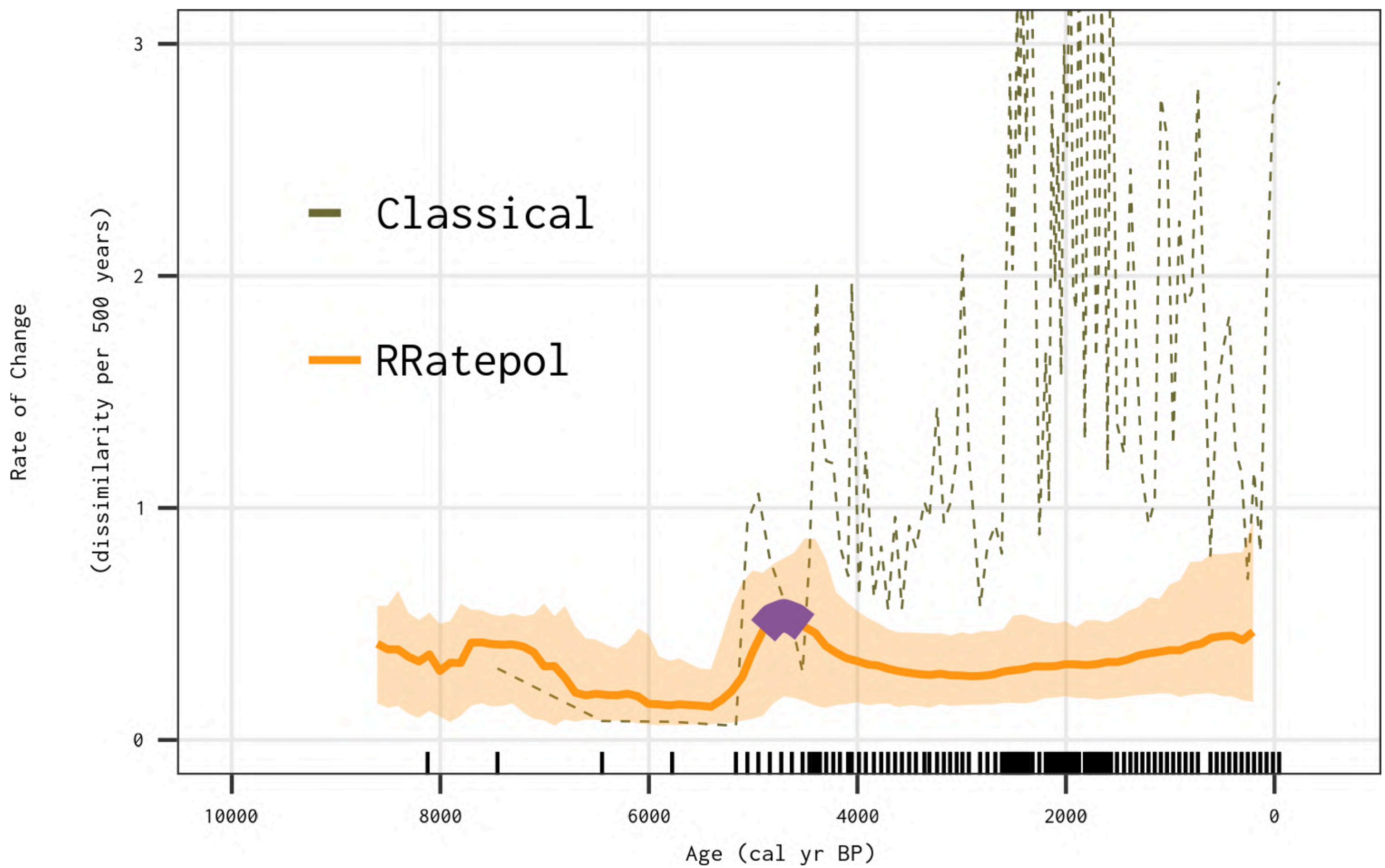
Mottl et al. (2021) *Review of Palaeobotany and Palynology*

{RRatepol}



Mottl et al. (2021) *Review of Palaeobotany and Palynology*

{RRatepol}



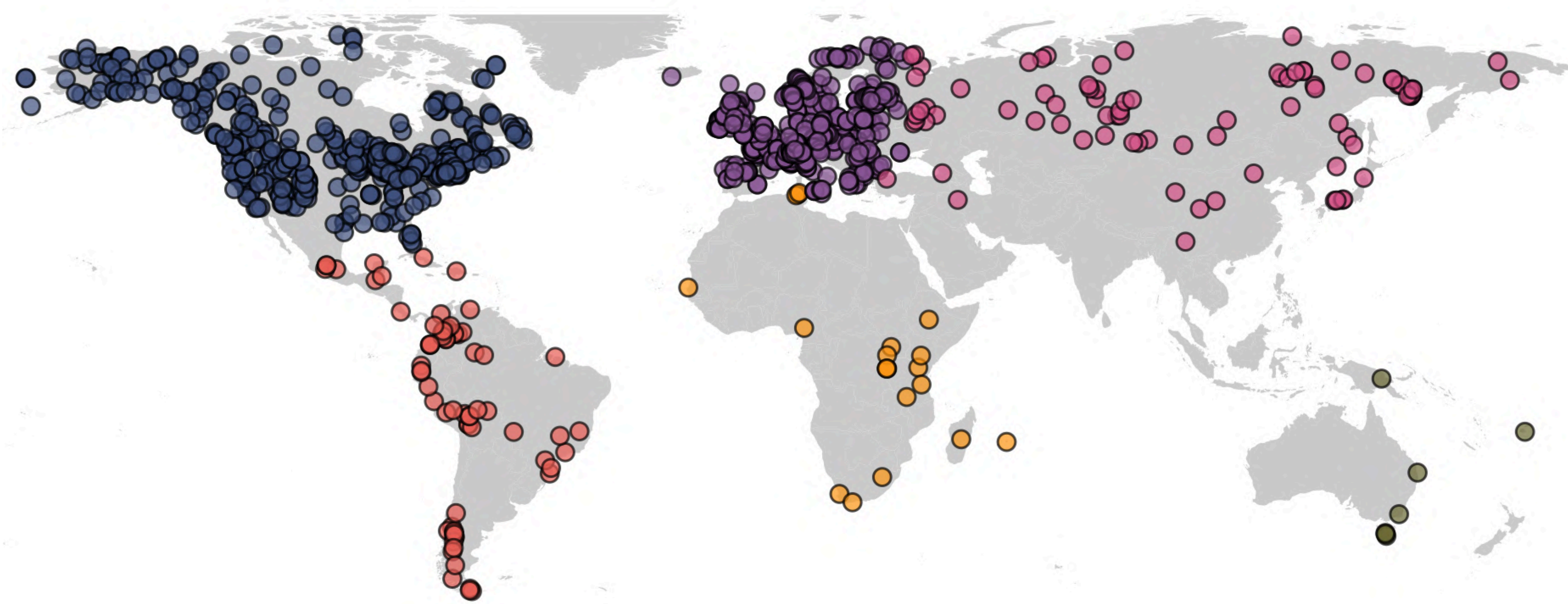
Mottl et al. (2021) *Review of Palaeobotany and Palynology*

Global rates of vegetation change over the past 18,000 years

Global assembly of fossil pollen records

Estimation of rate of vegetation change

Global rates of vegetation change

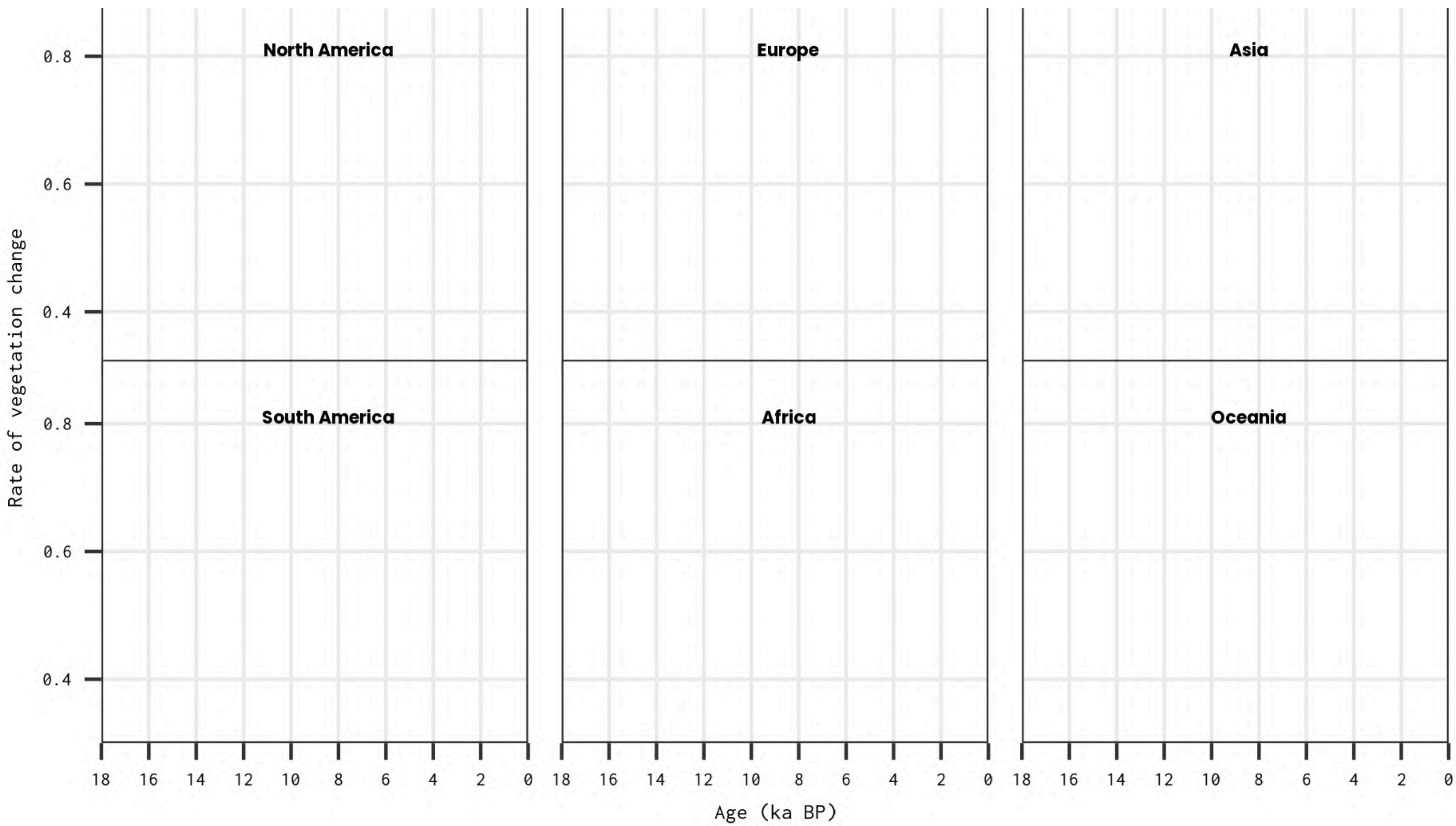


Over 1100 fossil pollen time series from all continents

(except Antarctica)

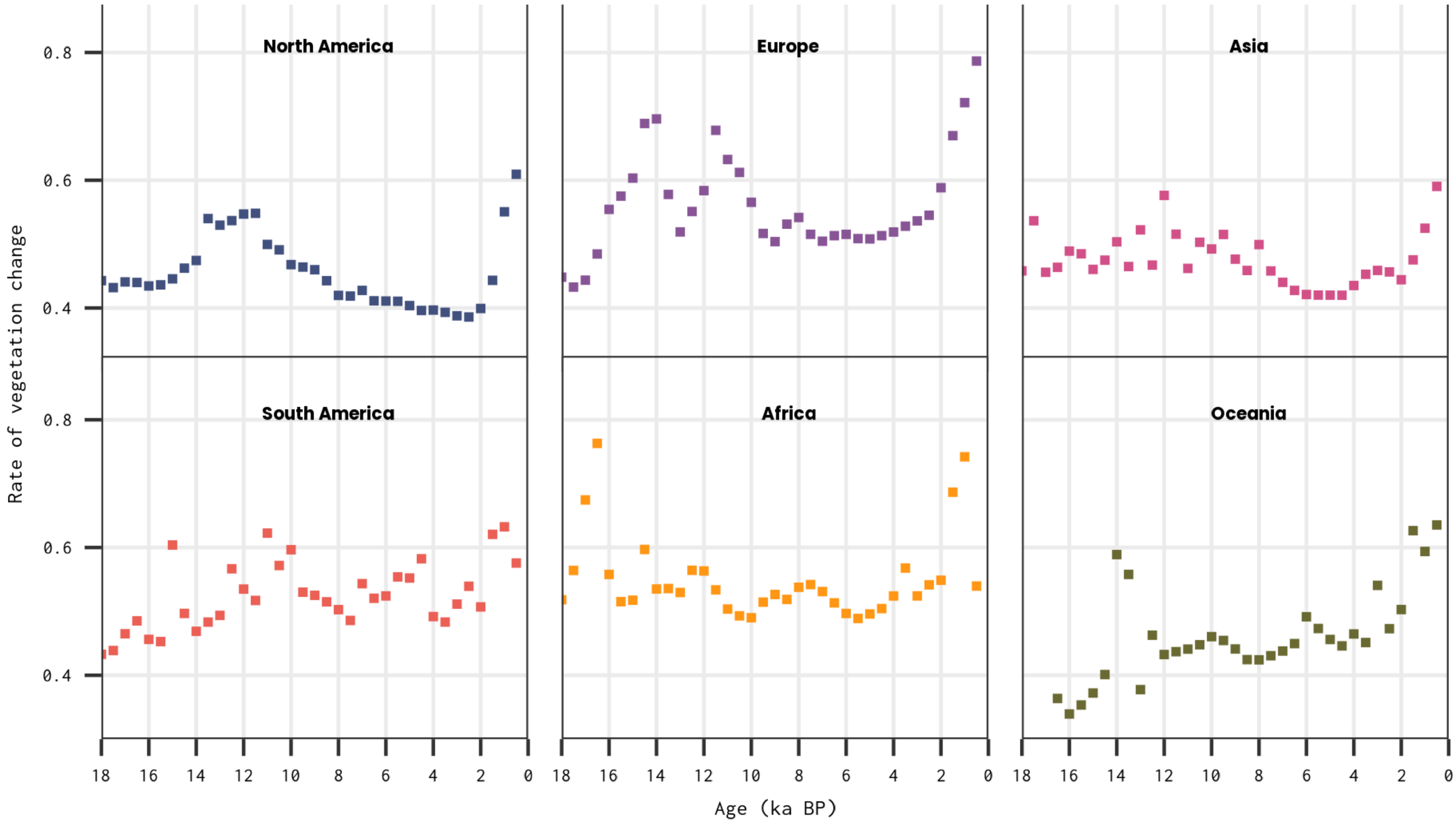
Adapted from Mottl & Flantua et al. (2021) *Science*

Global rates of vegetation change



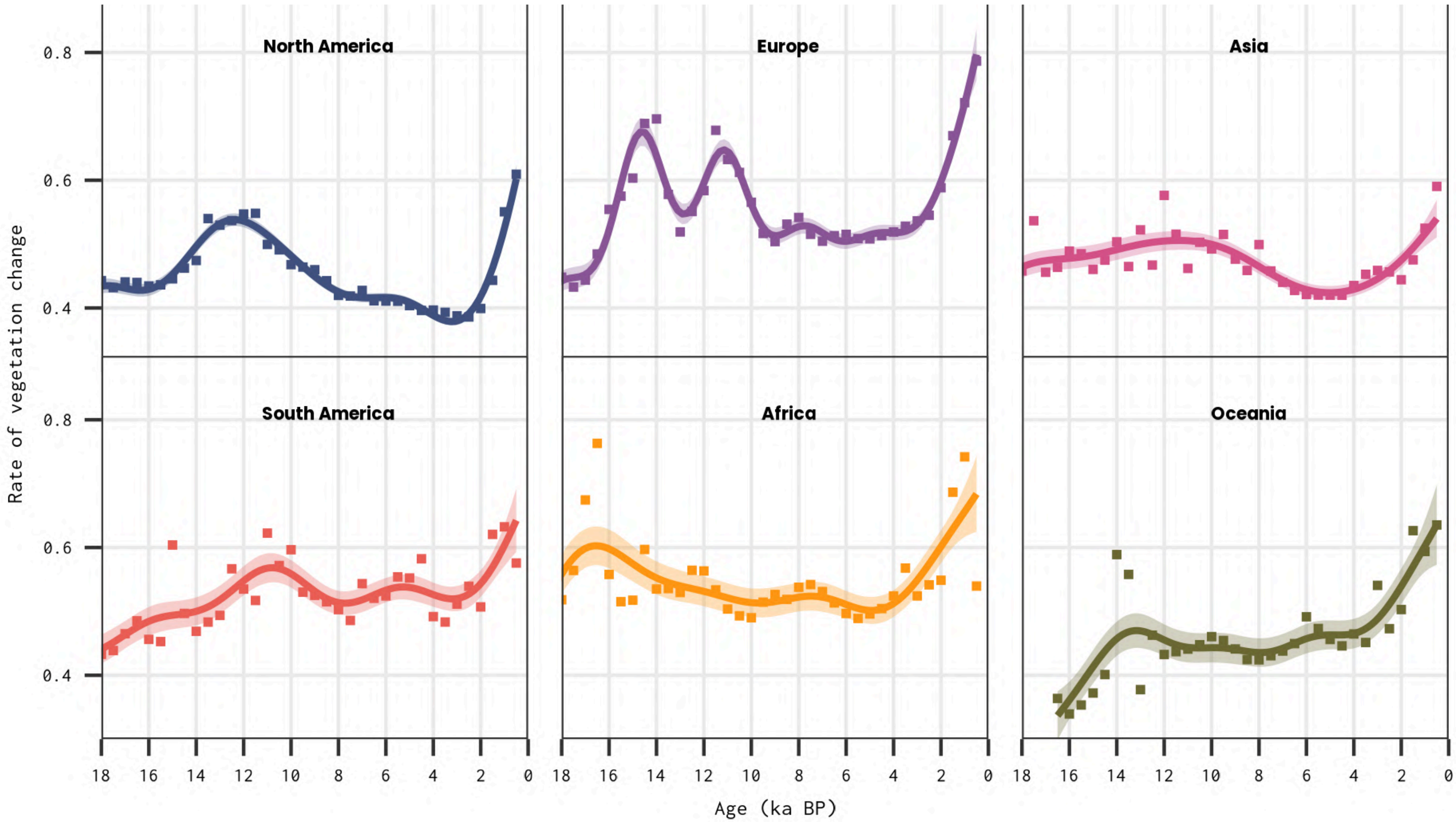
Adapted from Mottl & Flantua et al. (2021) *Science*

Global rates of vegetation change



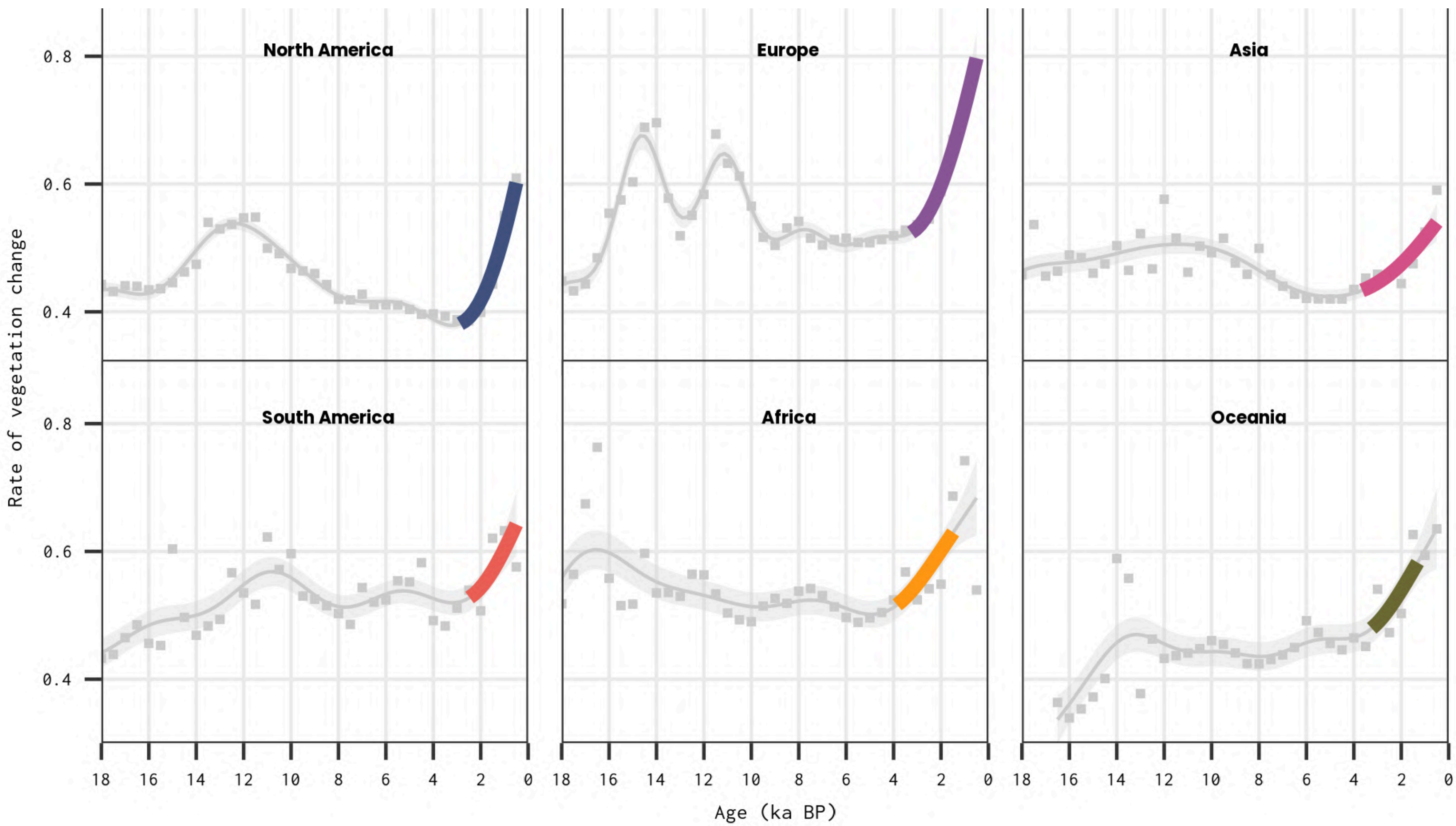
Adapted from Mottl & Flantua et al. (2021) *Science*

Global rates of vegetation change



Adapted from Mottl & Flantua et al. (2021) *Science*

Global rates of vegetation change



Adapted from Mottl & Flantua et al. (2021) *Science*

Science

HOME > SCIENCE > VOL. 372, NO. 6544 > GLOBAL ACCELERATION IN RATES OF VEGETATION CHANGE OVER THE PAST 18,000 YEARS

REPORT

Global acceleration in rates of vegetation change over the past 18,000 years

ONDŘEJ MOTTL, SUZETTE G. A. FLANTUA, KUBER P. BHATTA, VIVIAN A. FELDE, THOMAS GIESECKE, SIMON GORING, ERIC C. GRIMM, SIMON HABERLE, HENRY HOOGHIEMSTRA, SARAH IVORY, PETR KUNEŠ, STEFFEN WOLTERS, ALISTAIR W. R. SEDDON, AND JOHN W. WILLIAMS

SCIENCE • 21 May 2021 • Vol 372, Issue 6544 • pp. 860-864 • DOI: 10.1126/science.abg1685

DOI 10.1126.science.abg1685

Mottl & Flantua et al. (2021) *Science*

Continental Patterns

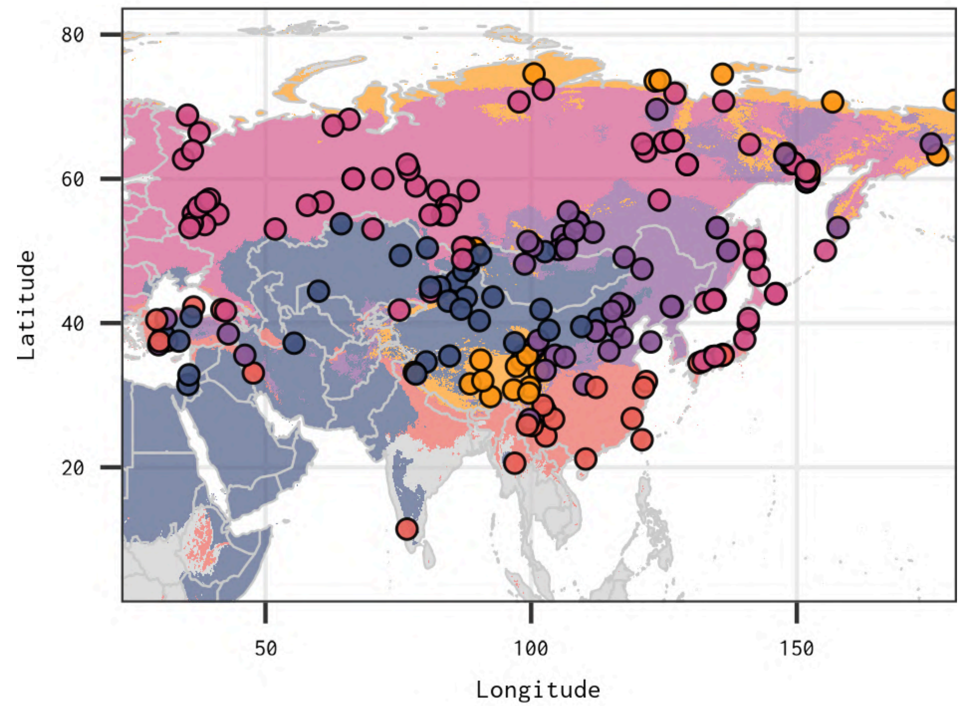
We do not have to stop at single ecosystem property (e.g., RoC)

Continental Patterns

We do not have to stop at single ecosystem property (e.g., RoC)

Asia as a case study spanning diverse climates and vegetation types

Asia Case Study



Adapted from Bhatta et al. (2023) *Frontiers in Ecology and Evolution*

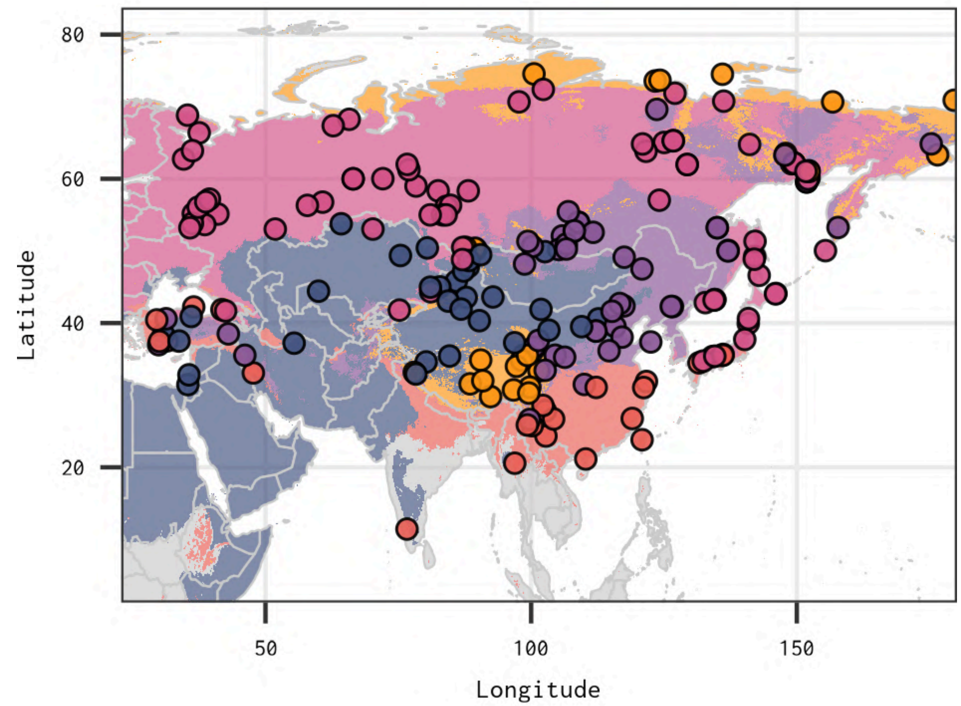
Continental Patterns

We do not have to stop at single ecosystem property (e.g., RoC)

Asia as a case study spanning diverse climates and vegetation types

Can we identify consistent continental trend of vegetation change across Asia during the Holocene?

Asia Case Study

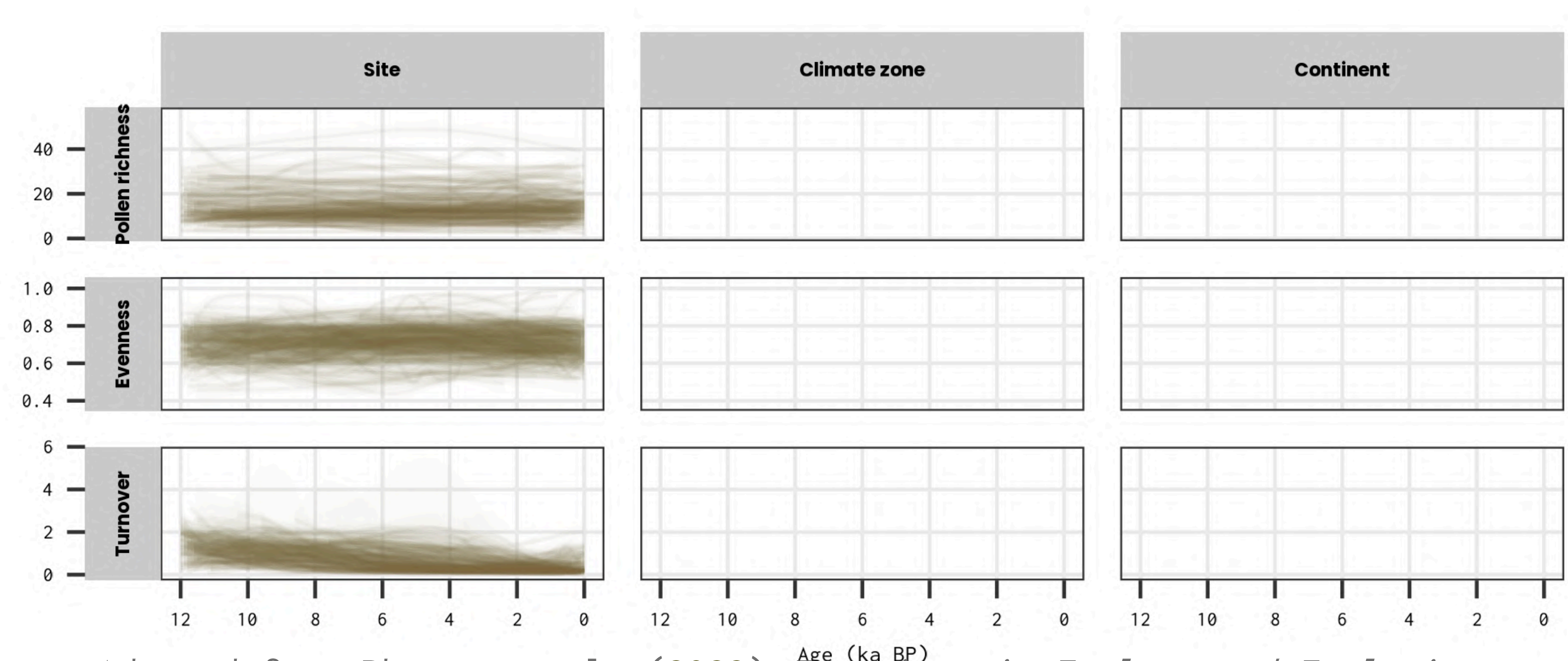


Adapted from Bhatta et al. (2023) *Frontiers in Ecology and Evolution*

frontiers

Exploring spatio-temporal patterns of palynological changes in Asia during the Holocene

Kuber P. Bhatta[1*†] Ondřej Mottl[1,2†] Vivian A. Felde[1,2†] Suzette G. A. Flantua[1,2†]
Hilary H. Birks[1,2†] Xianyong Cao[3†] Fahu Chen[3†] John-Arvid Grytnes[1†]
Alistair W. R. Seddon[1,2†] Harry John B. Birks[1,2,4†]

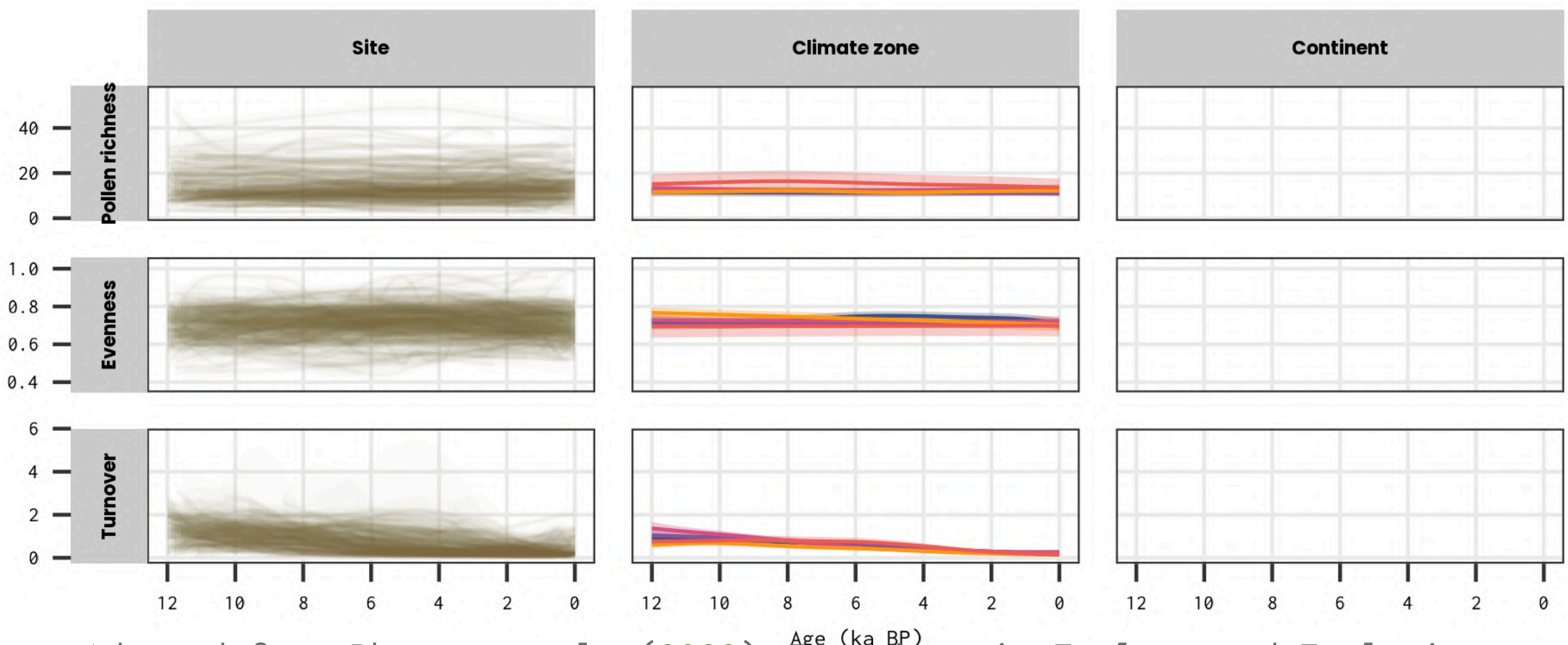


Adapted from Bhatta et al. (2023) *Frontiers in Ecology and Evolution*

frontiers

Exploring spatio-temporal patterns of palynological changes in Asia during the Holocene

Kuber P. Bhatta[1*†] Ondřej Mottl[1,2†] Vivian A. Felde[1,2†] Suzette G. A. Flantua[1,2†]
Hilary H. Birks[1,2†] Xianyong Cao[3†] Fahu Chen[3†] John-Arvid Grytnes[1†]
Alistair W. R. Seddon[1,2†] Harry John B. Birks[1,2,4†]

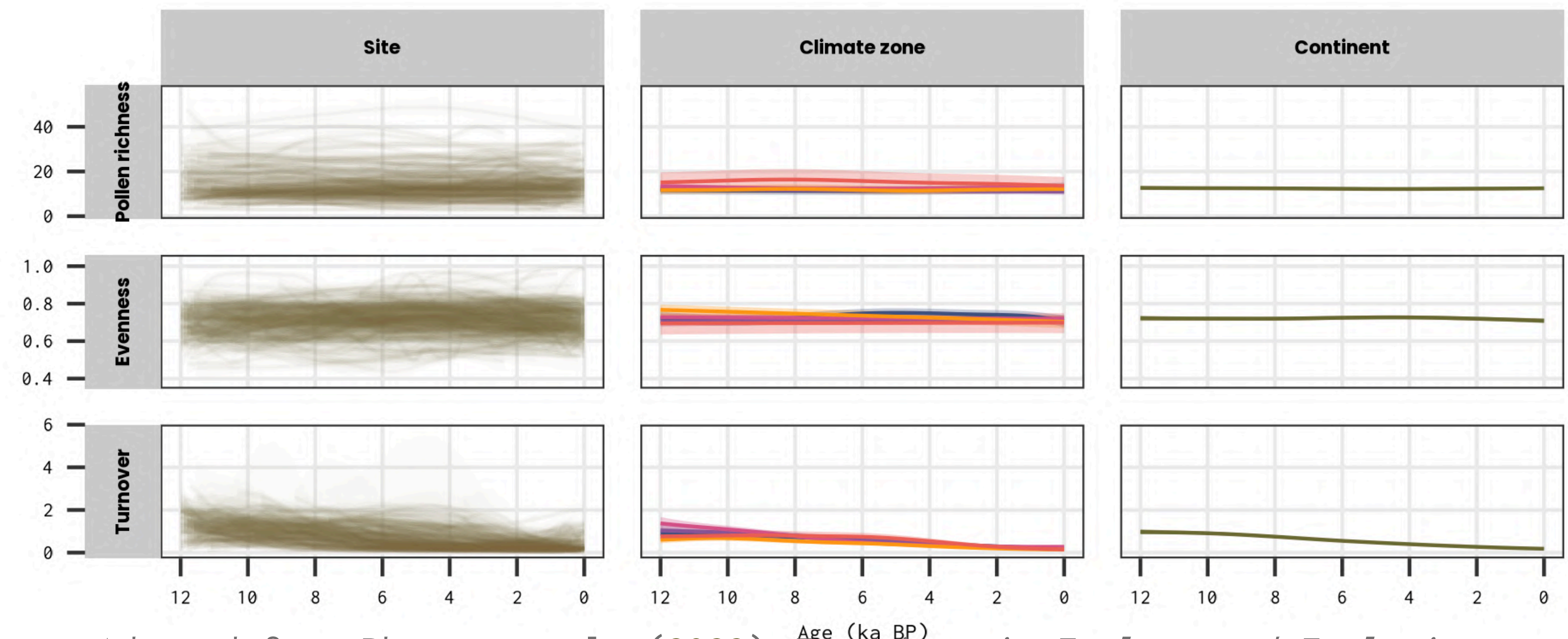


Adapted from Bhatta et al. (2023) *Frontiers in Ecology and Evolution*

frontiers

Exploring spatio-temporal patterns of palynological changes in Asia during the Holocene

Kuber P. Bhatta[1*†] Ondřej Mottl[1,2†] Vivian A. Felde[1,2†] Suzette G. A. Flantua[1,2†]
Hilary H. Birks[1,2†] Xianyong Cao[3†] Fahu Chen[3†] John-Arvid Grytnes[1†]
Alistair W. R. Seddon[1,2†] Harry John B. Birks[1,2,4†]



Adapted from Bhatta et al. (2023) *Frontiers in Ecology and Evolution*

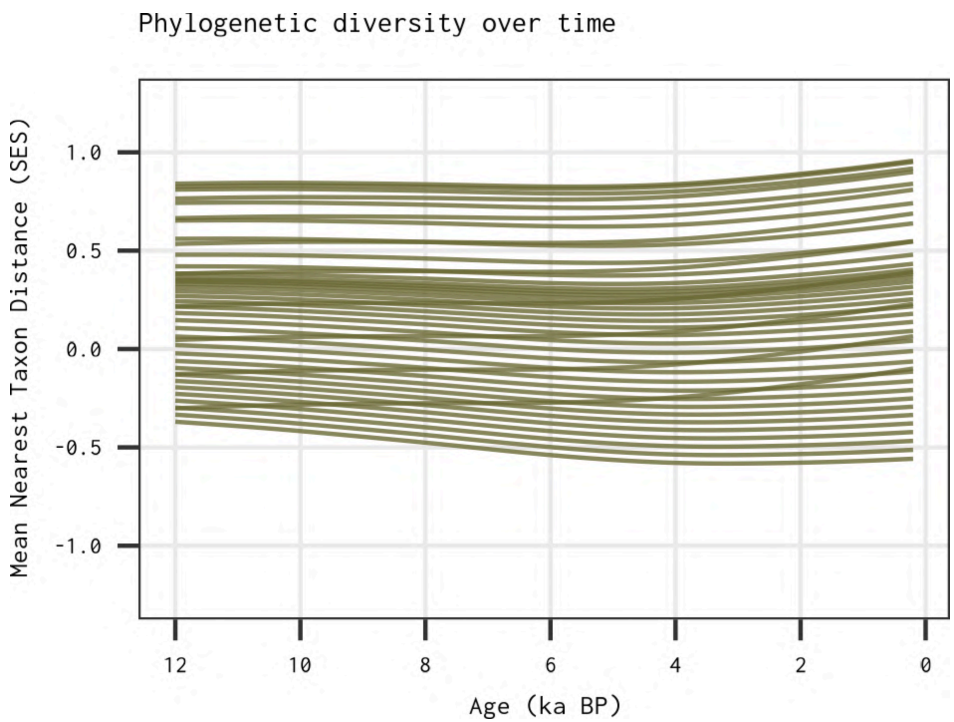
scientific reports

Article | Open access | Published: 02 August 2024

Latitudinal gradients in the phylogenetic assembly of angiosperms in Asia during the Holocene

Kuber P. Bhatta, Ondřej Mottl, Vivian A. Felde, John-Arvid Grytnes, Triin Reitalu, Hilary H. Birks, H. John B. Birks & Ole R. Vetaas

Scientific Reports **14**, Article number: 17940 (2024) | Cite this article



Adapted from Bhatta et al. (2024) *scientific reports*

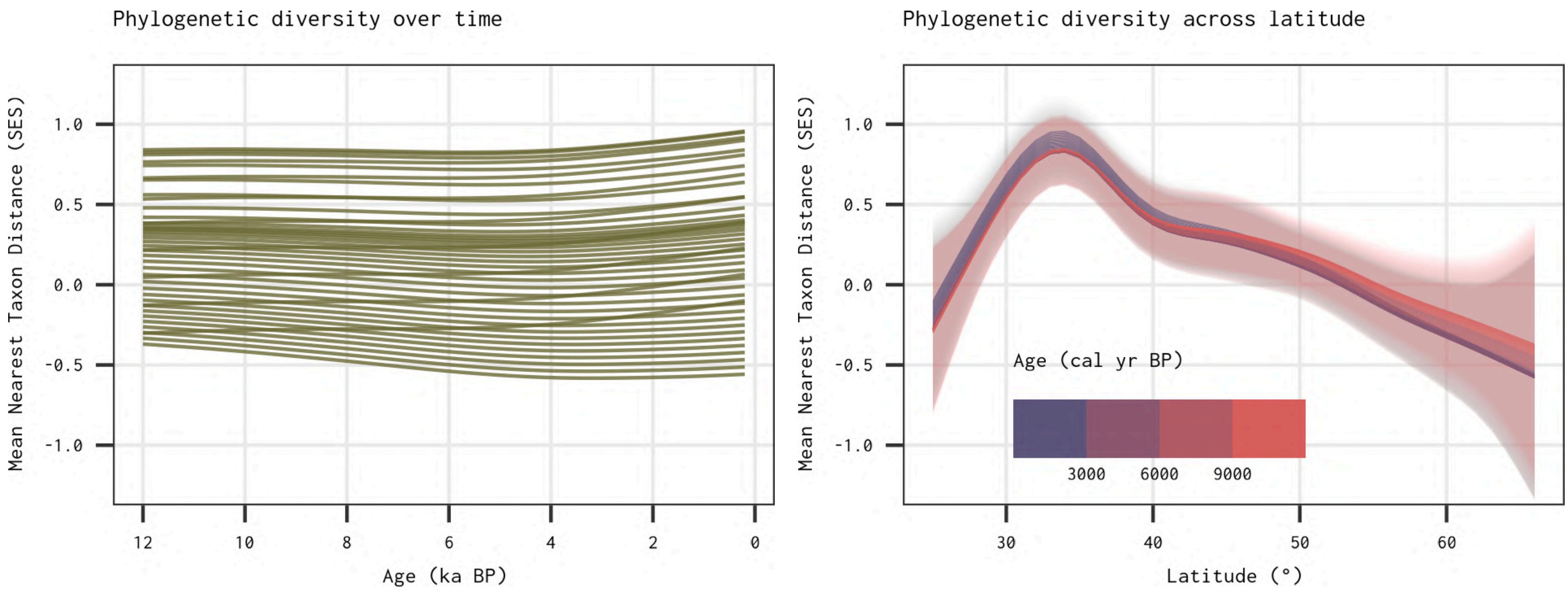
scientific reports

Article | Open access | Published: 02 August 2024

Latitudinal gradients in the phylogenetic assembly of angiosperms in Asia during the Holocene

Kuber P. Bhatta, Ondřej Mottl, Vivian A. Felde, John-Arvid Grytnes, Triin Reitalu, Hilary H. Birks, H. John B. Birks & Ole R. Vetaas

Scientific Reports **14**, Article number: 17940 (2024) | Cite this article

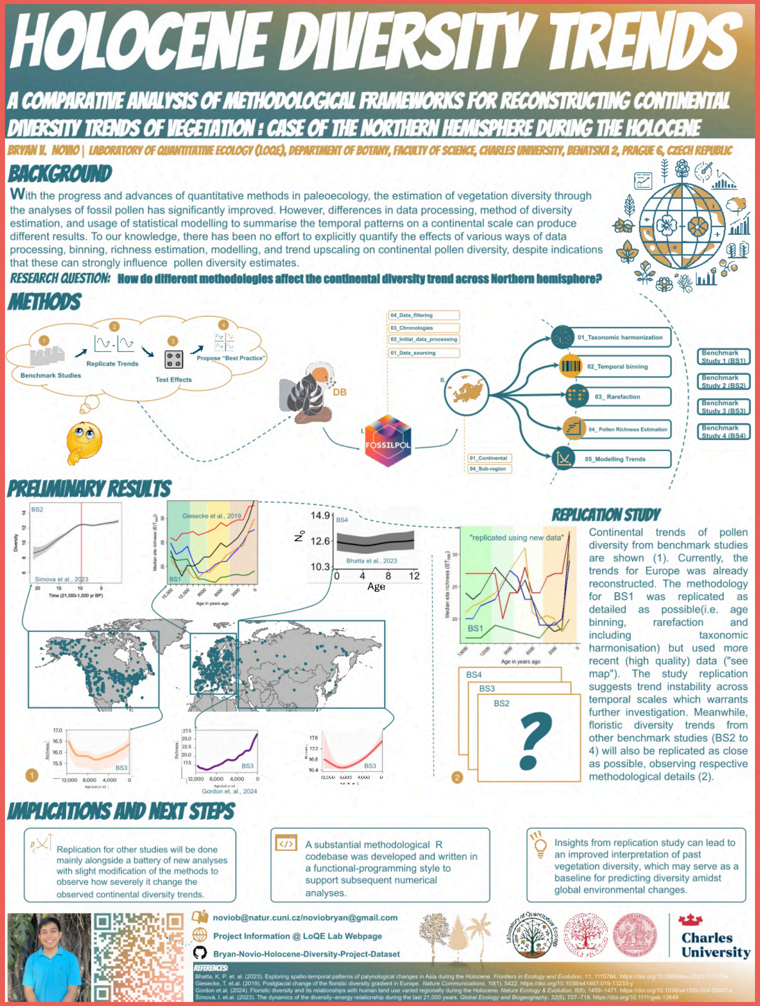


Adapted from Bhatta et al. (2024) *scientific reports*

Student intermezzo

Poster No. 2006

A Comparative Analysis of Methodological Frameworks for Reconstructing Continental Diversity Trends of Vegetation



Functional Traits

- Taxonomic identity ≠ ecological function
- Functional ecology allow to generalise across taxa, regions, and time
- Link between palaeoecological proxies and functional traits

Figure 1 from Brown et al. (2023) *Functional Ecology*

Functional Traits

- Taxonomic identity ≠ ecological function
- Functional ecology allow to generalise across taxa, regions, and time
- Link between palaeoecological proxies and functional traits

BRITISH ECOLOGICAL SOCIETY Functional Ecology

REVIEW | Open Access

Trait-based approaches as ecological time machines: Developing tools for reconstructing long-term variation in ecosystems

Kerry A. Brown, M. Jane Bunting, Fabio Carvalho, Francesco de Bello, Luke Mander, Katarzyna Marcisz, Ondrej Mottl, Triin Reitalu, Jens-Christian Svenning

First published: 22 August 2023 | https://doi.org/10.1111/1365-2435.14415 |

VIEW METRICS

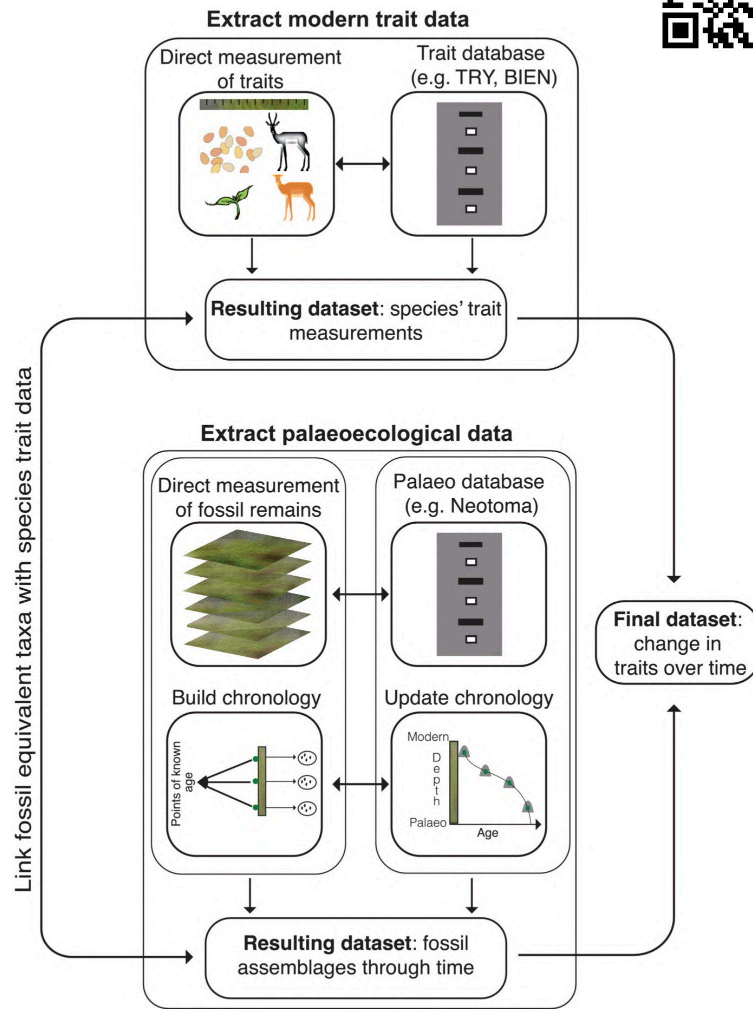


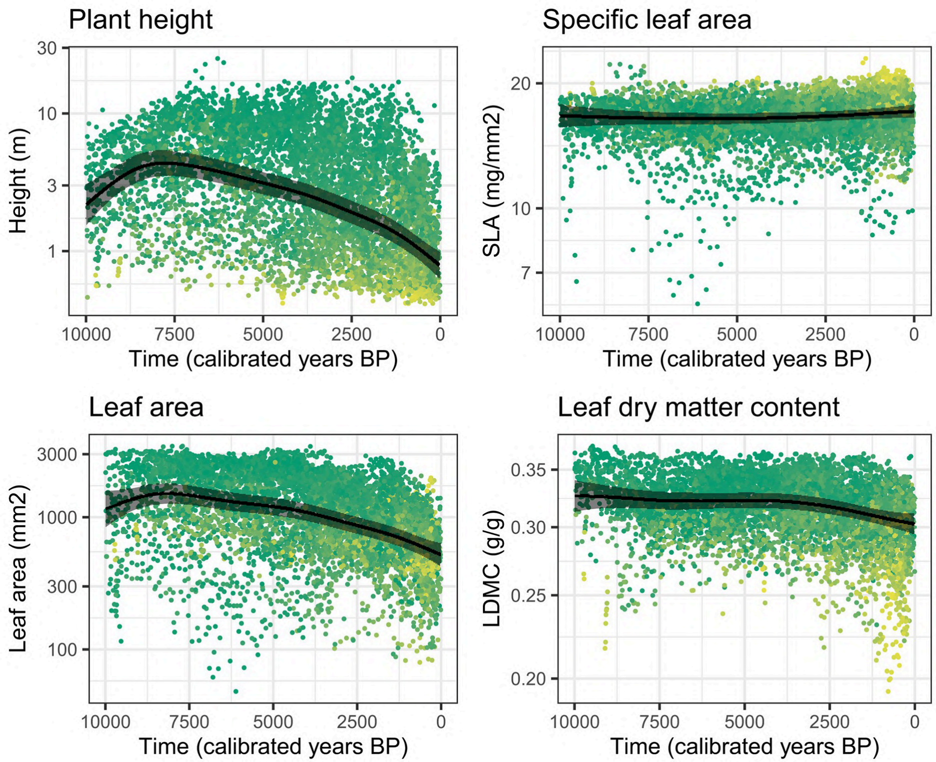
Figure 1 from Brown et al. (2023) *Functional Ecology*

Student intermezzo

Poster No. 2005

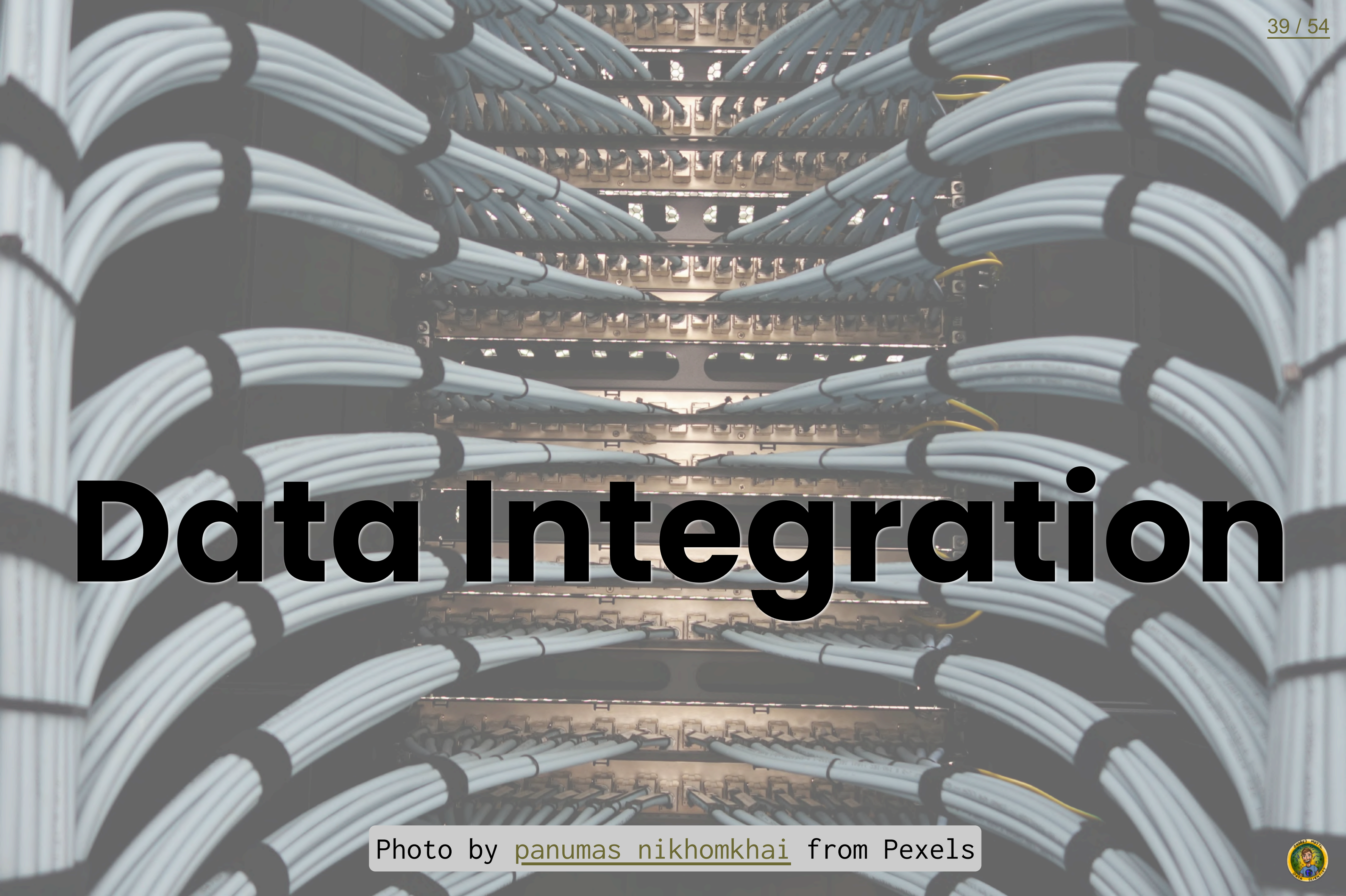
Functional Vegetation Paleoecology: Mitigate the impact of climate change with data-driven insight from past vegetation

Traits in time



Example for subcontinental scale integration of functional traits with fossil pollen data

Figure 4 from Veeken et al. (2022) *Ecology Letters*



Data Integration

Photo by panumas nikhomkhai from Pexels

Denmark example

*How has vegetation in **Denmark** remained **functionally stable** under **climate** change from the **LGM to the present**?*

Denmark example

*How has vegetation in **Denmark** remained **functionally stable** under **climate** change from the **LGM to the present**?*

🎯 Data Needs

- Fossil pollen and contemporary vegetation plots
- Functional traits (e.g. height, SLA, seed mass)
- Climate data (past and contemporary)

Denmark example

*How has vegetation in **Denmark** remained **functionally stable** under **climate** change from the **LGM to the present**?*

🎯 Data Needs

- Fossil pollen and contemporary vegetation plots
- Functional traits (e.g. height, SLA, seed mass)
- Climate data (past and contemporary)

🚧 Obstacles

- ❌ Several data sources (formats, tools),
- ❌ Taxonomies mismatch across datasets
- ❌ Technically challenging (with limited reproducibility)

VegVault + {vaultkeepr}

How has vegetation in ***Denmark*** *remain* ***functionally stable*** *under* ***climate*** *change from the* ***LGM to the present****?*

With few lines of code, you can get the data you need to answer that!

```r
data_denmark <-
  vaultkeepr::open_vault(
    path = "<path_to_VegVault>"
  ) |>
  vaultkeepr::get_datasets() |>
  vaultkeepr::select_dataset_by_geo(
    lat_lim = c(54.5, 58.0),
    long_lim = c(8.0, 13.0)
  ) |>
  vaultkeepr::get_samples() |>
  vaultkeepr::select_samples_by_age(
    age_lim = c(0, 20000) # 20,000 BP
  ) |>
  vaultkeepr::get_taxa(
    classify_to = "genus"
  ) |>
  vaultkeepr::get_abiotic_data() |>
  vaultkeepr::get_traits(
    classify_to = "genus"
  ) |>
  vaultkeepr::extract_data()
```

VegVault database

scientific **data**

nature > scientific data > data descriptors > article

Data Descriptor | Open access | Published: 05 December 2025

VegVault dataset: linking global paleo-, and neo-vegetation data with functional traits and abiotic drivers

Ondřej Mottl, Franka Gaiser, Irena Šímová & Suzette G. A. Flantua

- A global SQLite database linking interdisciplinary plot-based vegetation data!
- Links to functional traits, climate variables, and soil properties
- Built for exploring **biodiversity** dynamics across **time & space**
- Open Source & Reproducible
- 🔗 Learn more: bit.ly/VegVault

Mottl et al. (2025) *Scientific Data*

VegVault database

~110 GB | 32 tables & 87 variables | 480,000+ Datasets | 13M+ Samples | 110,000+ Taxa | 11M+ Trait values | 8 Abiotic variables

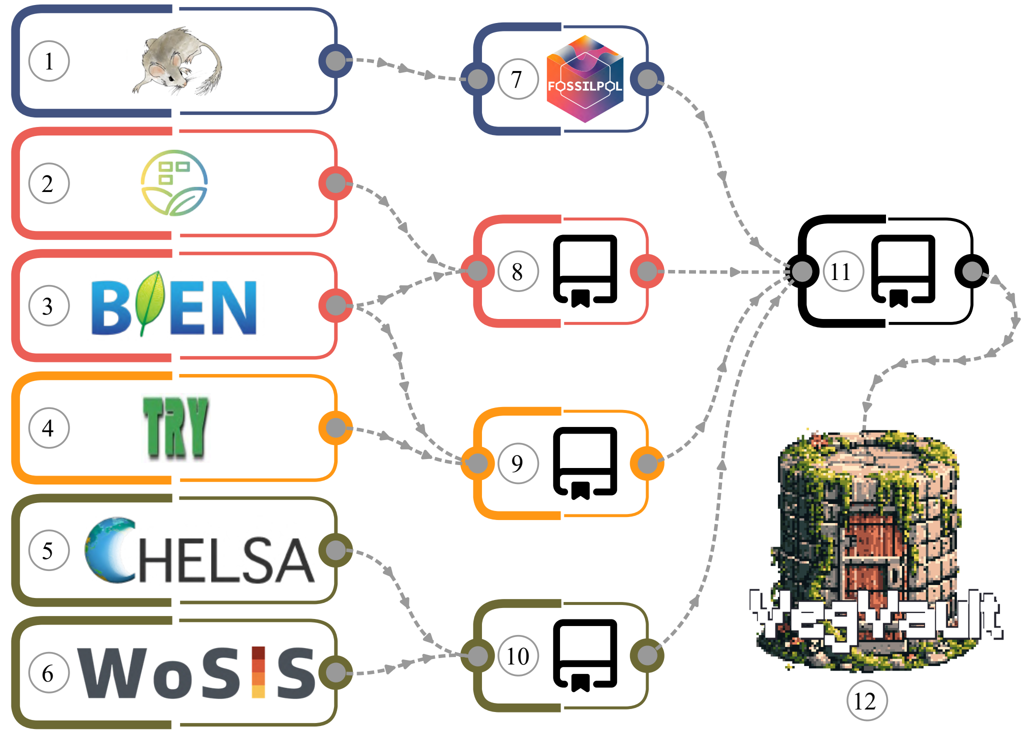
Mottl et al. (2025) *Scientific Data*

VegVault database

~110 GB | 32 tables & 87 variables | 480,000+ Datasets | 13M+ Samples | 110,000+ Taxa | 11M+ Trait values | 8 Abiotic variables

Data sources

- 🌿 Paleoecological records (*Neotoma*, *FOSSILPOL*)
- 🌱 Contemporary vegetation (*sPlotOpen*, *BIEN*)
- 🌾 Functional traits (*BIEN*, *TRY*)
- 🌍 Climate & soil data (*CHELSA*, *WoSIS*)



Mottl et al. (2025) *Scientific Data*

{vaultkeepr} package

An Open Source R package 📦 - Your key 🔑 to **VegVault**

- ✅ No SQL required - just load & go!
- ✅ Filter by taxa, traits, time & space
- ✅ Tidyverse-compatible interface
- ✅ Directly usable in R for analysis
- 🔗 Learn more: bit.ly/vaultkeepr

R-CMD-check passing
codecov 98%
CRAN not published

Mottl et al. (2025) *Scientific Data*

VegVault database

💎 Key Innovations

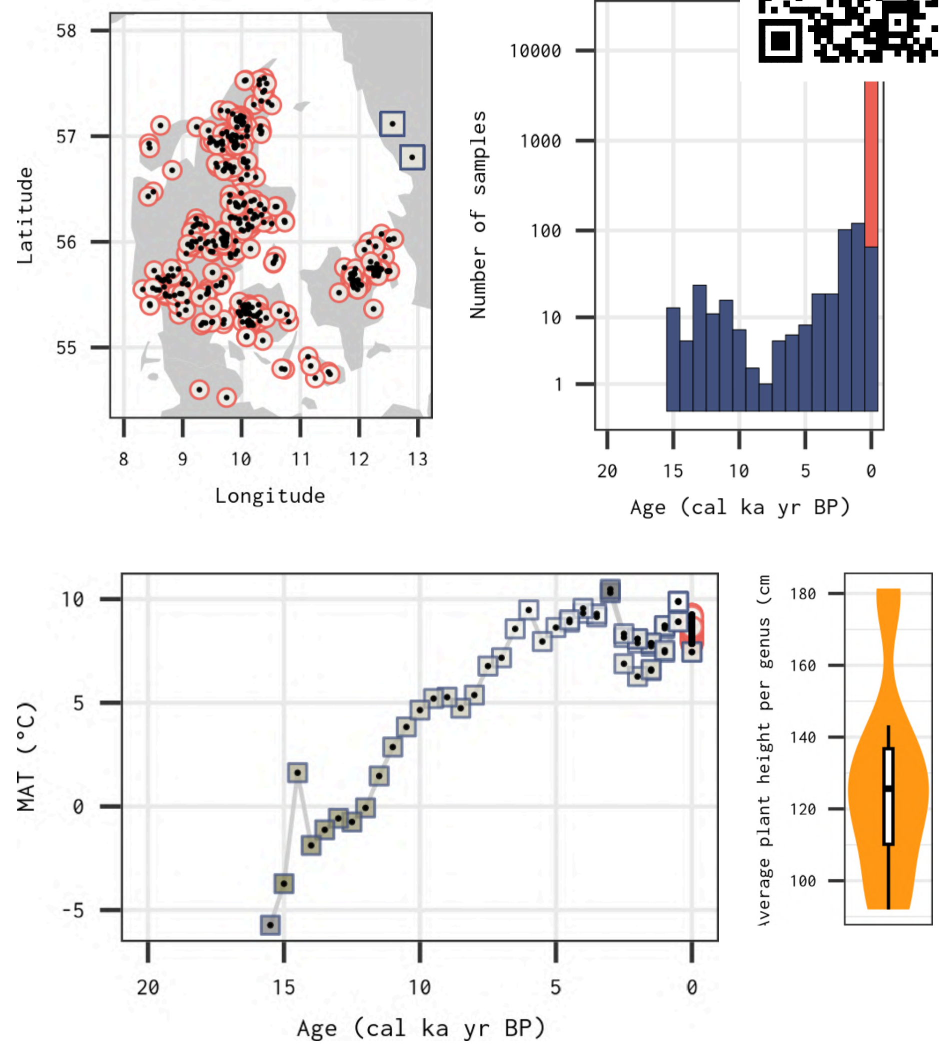
- Integration of interdisciplinary vegetation data into a single database
- Linking vegetation data with functional traits and abiotic variables
- User-friendly R package {vaultkeepr} for data access and extraction
- Open Source & Reproducible workflows for vegetation research

Mottl et al. (2025) *Scientific Data*

VegVault database

💎 Key Innovations

- Integration of interdisciplinary vegetation data into a single database
- Linking vegetation data with functional traits and abiotic variables
- User-friendly R package {vaultkeepr} for data access and extraction
- Open Source & Reproducible workflows for vegetation research

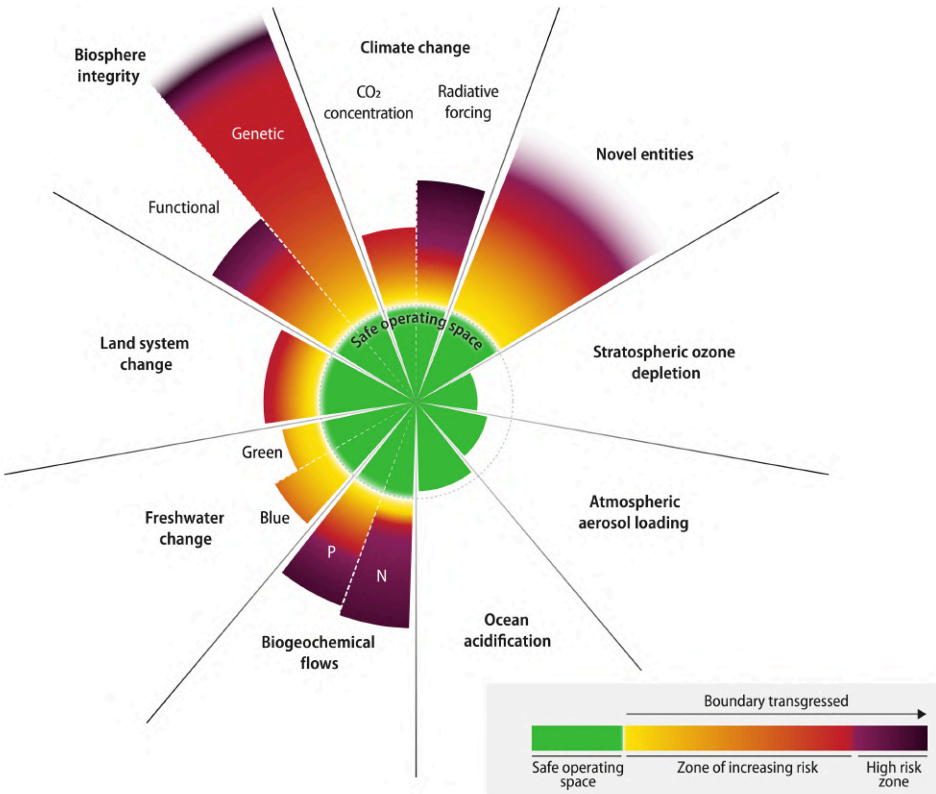


Mottl et al. (2025) *Scientific Data*

Planetary Boundaries

Photo by Markus Spiske from Pexels

Planetary Boundaries



ScienceAdvances

HOME > SCIENCE ADVANCES > VOL. 9, NO. 37 > EARTH BEYOND SIX OF NINE PLANETARY BOUNDARIES

Earth beyond six of nine planetary boundaries

KATHERINE RICHARDSON, WILL STEFFEN, WOLFGANG LUCHT, JØRGEN BENDTSEN, SARAH E. CORNELL, JONATHAN F. DONGES, MARKUS DRÜKE, INGO FETZER, GOVINDASAMY BALA, [...], AND JOHAN ROCKSTRÖM +19 authors Authors Info & Affiliations

SCIENCE ADVANCES • 13 Sep 2023 • Vol 9, Issue 37 • DOI: 10.1126/sciadv.adh2458

Richardson et al. (2023) *Science Advances*

From Paleo to Policy

Global Change Biology

REVIEW | Open Access

Exploring the Interface Between Planetary Boundaries and Palaeoecology

This article relates to:

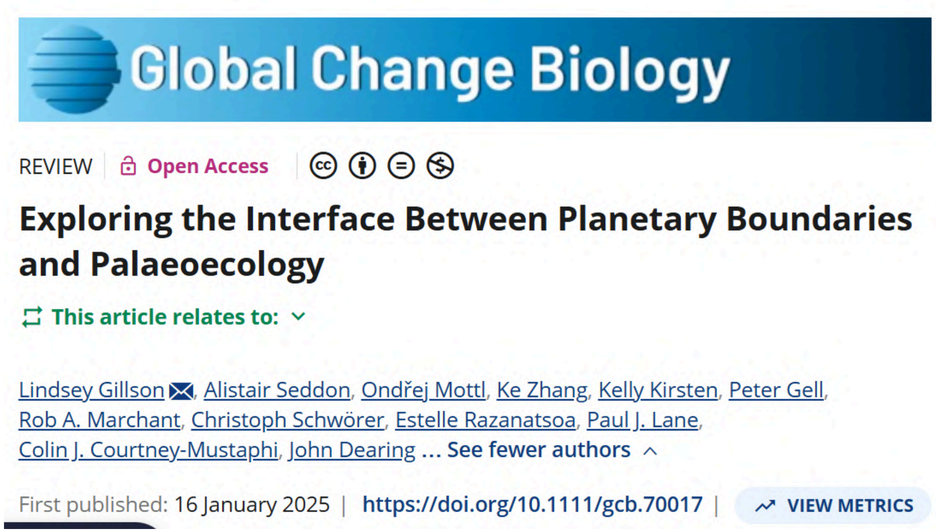
Lindsey Gillson, Alistair Seddon, Ondřej Mottl, Ke Zhang, Kelly Kirsten, Peter Gell, Rob A. Marchant, Christoph Schwörer, Estelle Razanatsoa, Paul J. Lane, Colin J. Courtney-Mustaphi, John Dearing ... See fewer authors

First published: 16 January 2025 | https://doi.org/10.1111/gcb.70017 | VIEW METRICS

Gillson et al. (2025) *Global Change Biology*

From Paleo to Policy

- Paleoecology offers long-term baselines for historical ranges of variability


Global Change Biology

REVIEW | Open Access

Exploring the Interface Between Planetary Boundaries and Palaeoecology

This article relates to:

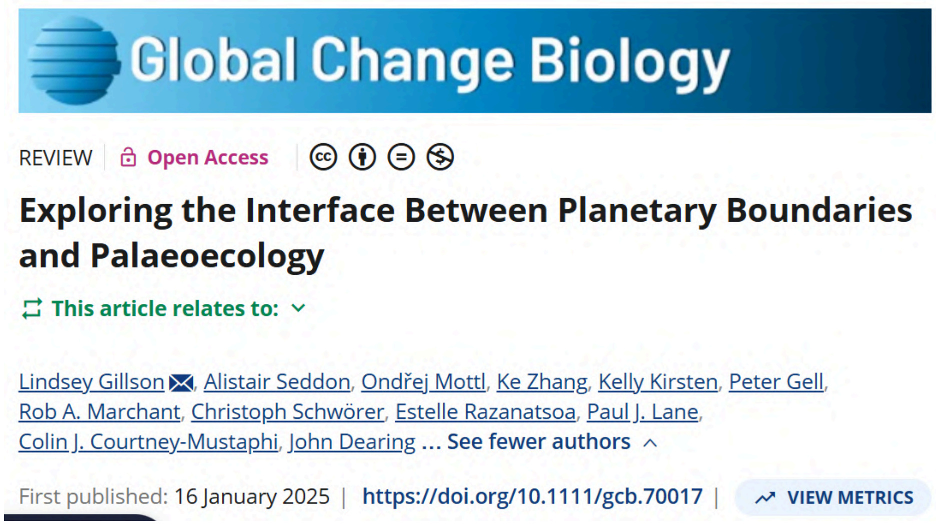
Lindsey Gillson, Alistair Seddon, Ondřej Mottl, Ke Zhang, Kelly Kirsten, Peter Gell, Rob A. Marchant, Christoph Schwörer, Estelle Razanatsoa, Paul J. Lane, Colin J. Courtney-Mustaphi, John Dearing ... See fewer authors

First published: 16 January 2025 | https://doi.org/10.1111/gcb.70017 | VIEW METRICS

Gillson et al. (2025) *Global Change Biology*

From Paleo to Policy

- Paleoecology offers long-term baselines for historical ranges of variability
- RAD (resist/accept/direct) framework for integrating paleoecological data into policy and decision-making



Global Change Biology

REVIEW | Open Access

Exploring the Interface Between Planetary Boundaries and Palaeoecology

This article relates to:

Lindsey Gillson, Alistair Seddon, Ondřej Mottl, Ke Zhang, Kelly Kirsten, Peter Gell, Rob A. Marchant, Christoph Schwörer, Estelle Razanatsoa, Paul J. Lane, Colin J. Courtney-Mustaphi, John Dearing ... See fewer authors

First published: 16 January 2025 | https://doi.org/10.1111/gcb.70017 | VIEW METRICS

Gillson et al. (2025) *Global Change Biology*

From Paleo to Policy

Global Change Biology

REVIEW | Open Access

Exploring the Interface Between Planetary Boundaries and Palaeoecology

This article relates to:

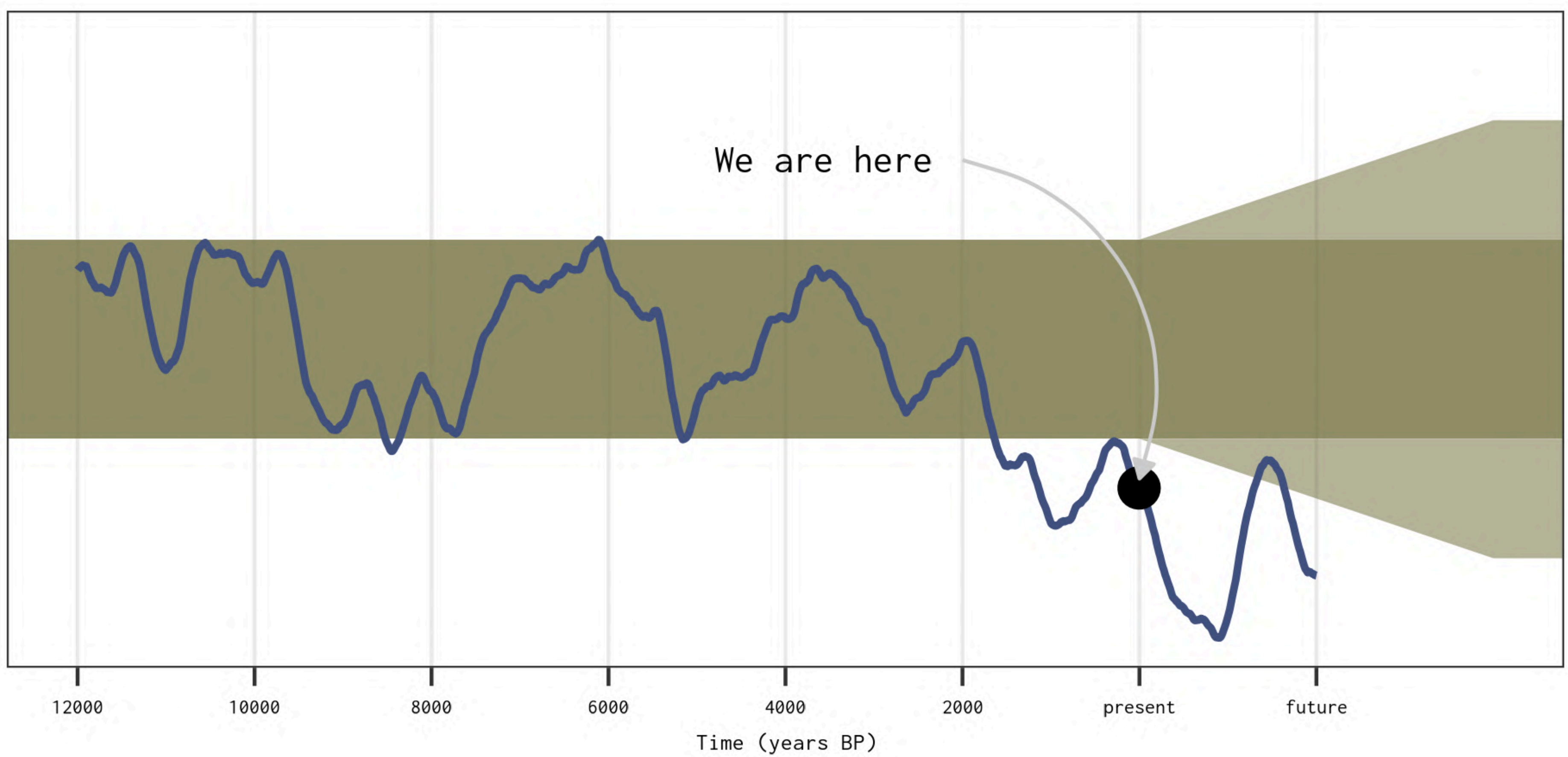
Lindsey Gillson, Alistair Seddon, Ondřej Mottl, Ke Zhang, Kelly Kirsten, Peter Gell, Rob A. Marchant, Christoph Schwörer, Estelle Razanatsoa, Paul J. Lane, Colin J. Courtney-Mustaphi, John Dearing ... See fewer authors

First published: 16 January 2025 | https://doi.org/10.1111/gcb.70017 | VIEW METRICS

- Paleoecology offers long-term baselines for historical ranges of variability
- RAD (resist/accept/direct) framework for integrating paleoecological data into policy and decision-making
- Even if the past is an imperfect analogue, it is still essential for bounding plausible variability and diagnosing novelty.

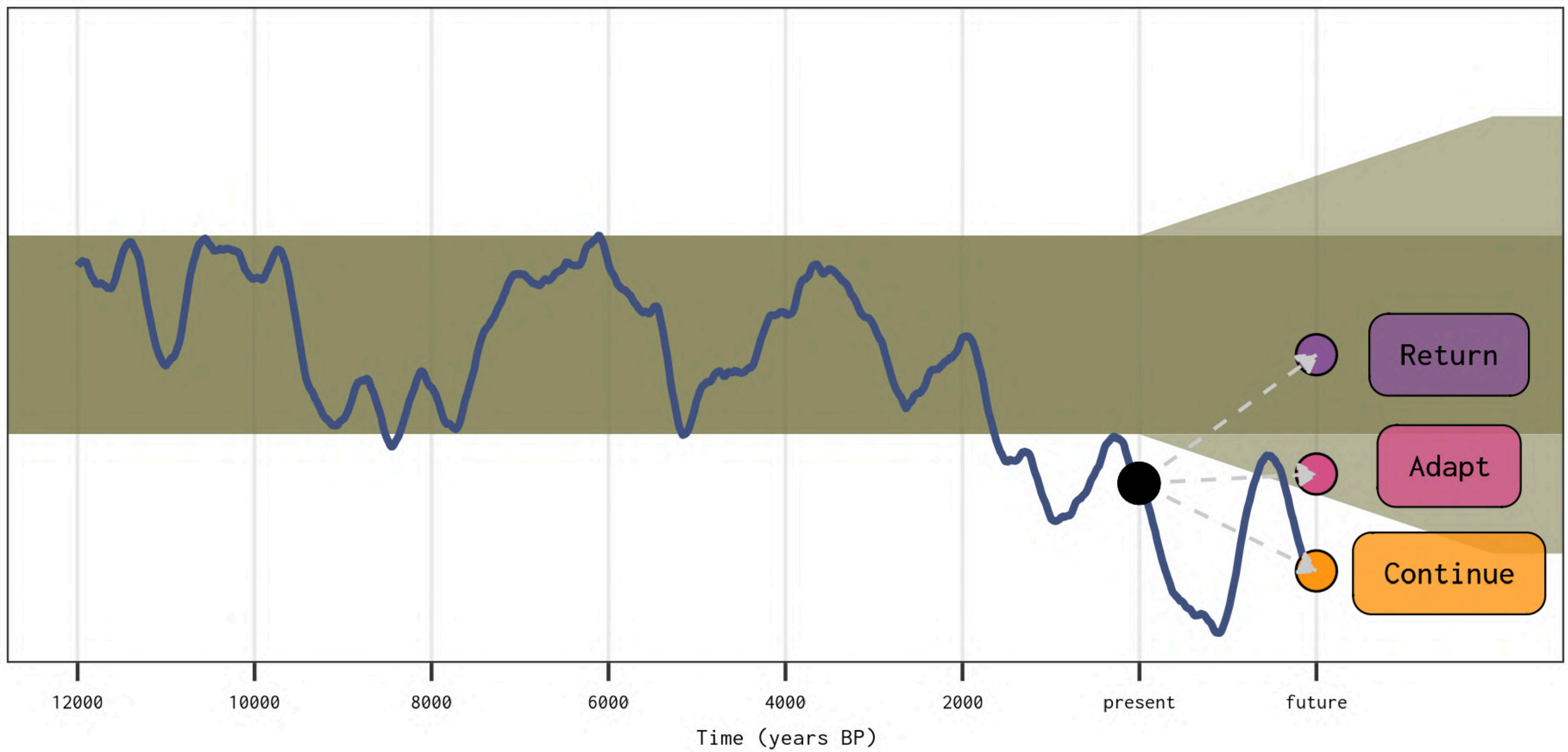
Gillson et al. (2025) *Global Change Biology*

From Paleo to Policy



Gillson et al. (2025) *Global Change Biology*

From Paleo to Policy



Gillson et al. (2025) *Global Change Biology*

The time is ripe … 👍

Photo by Felix Mittermeier from Pexels

The time is ripe … 👍

- Global (integrated) ecological databases spanning large spatio-temporal scales

Photo by Felix Mittermeier from Pexels

The time is ripe … 👍

- Global (integrated) ecological databases spanning large spatio-temporal scales
- Reproducible workflows for data access

Photo by Felix Mittermeier from Pexels

The time is ripe … 👍

- Global (integrated) ecological databases spanning large spatio-temporal scales
- Reproducible workflows for data access
- Ways to study various facets of biodiversity (taxonomic, functional, phylogenetic)

Photo by Felix Mittermeier from Pexels

The time is ripe … 👍

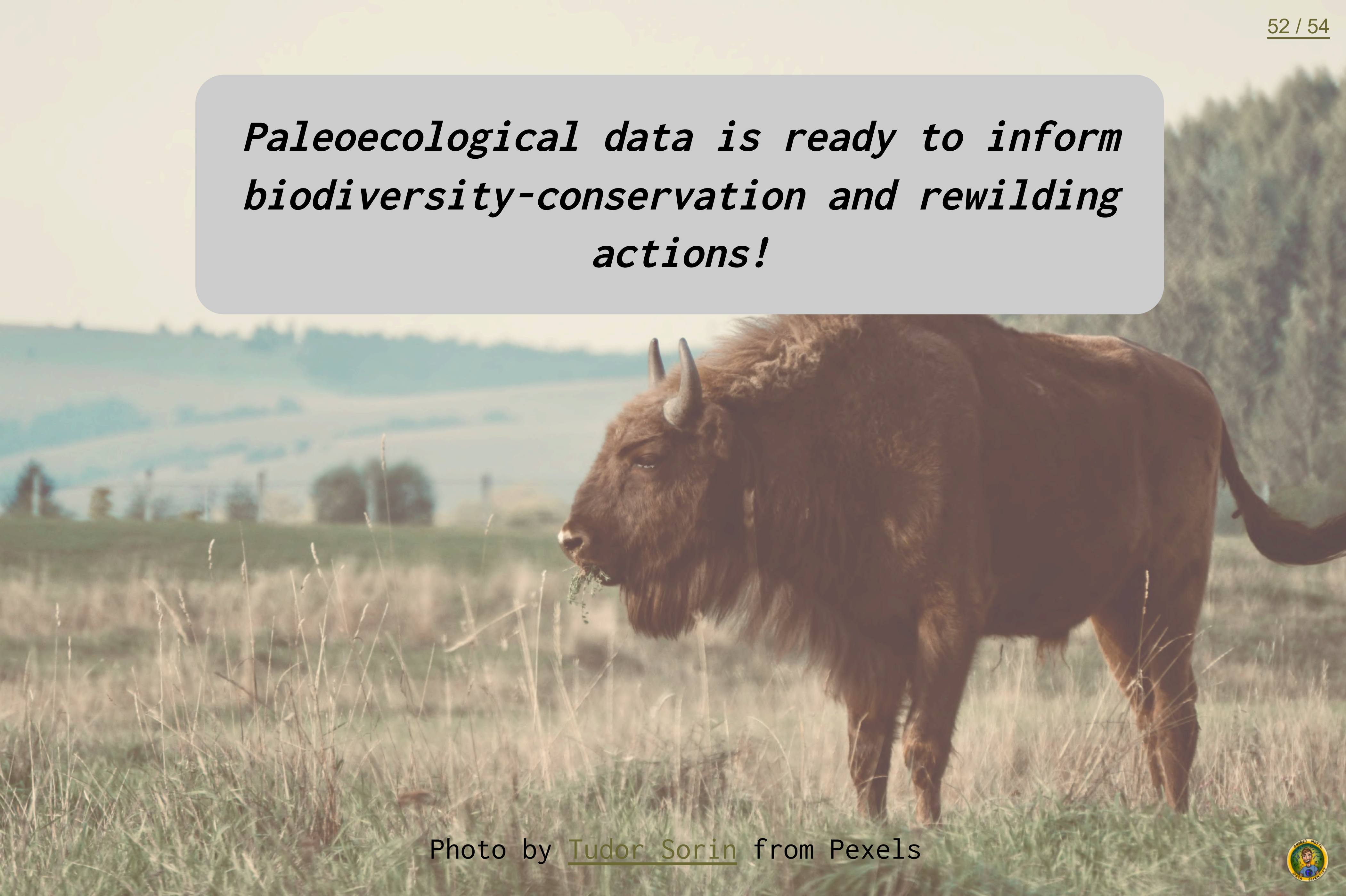
- Global (integrated) ecological databases spanning large spatio-temporal scales
- Reproducible workflows for data access
- Ways to study various facets of biodiversity (taxonomic, functional, phylogenetic)
- Statistical methods to handle complex hierarchical data structures

Photo by Felix Mittermeier from Pexels

The time is ripe ... 👍

- Global (integrated) ecological databases spanning large spatio-temporal scales
- Reproducible workflows for data access
- Ways to study various facets of biodiversity (taxonomic, functional, phylogenetic)
- Statistical methods to handle complex hierarchical data structures
- Computational power and Open Source tools for uncertainty-incorporating analysis

Photo by Felix Mittermeier from Pexels



Paleoecological data is ready to inform biodiversity-conservation and rewilding actions!

Photo by Tudor Sorin from Pexels

This presentation

is publically available!

🐱 Code | 🖼️ Slides

Estimating Vegetation Change from Fossil Pollen to Planetary Boundaries

A Global, Long-Term Synthesis Using Big Data

- 📧 ondrej.mottl(at)natur.cuni.cz
- 🦋 ondrejmottl.bsky.social
- 🐱 Github
- ID ORCID
- 🌐 Personal webpage

👤 Ondřej Mottl

TIBS 2026

CHARLES UNIVERSITY

My trip has been funded by Czech Science Foundation PIF grant (GN23-06386I)